MECWIN®

EMPOWERING FUTURE

STANDARD BLDC MOTORS

PLANETARY & WORM GEARED MOTORS

BRUSHLESS DC MOTORS ·

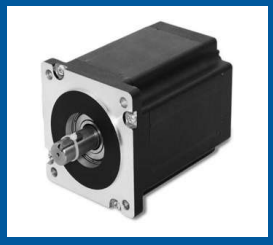
- 10 W to 15 kW output
- 12 V to 310 V DC supply
- 28 mm to 132 mm frame sizes
- Planetary & RV worm gearboxes
- Encoder and brake options
- Built in Bengaluru, India













MECWIN TECHNOLOGIES INDIA PVT. LTD.

Bengaluru,India · Engineeringoffice:Rugby,UnitedKingdom

www.mecwinindia.com

marketing@mecwinindia.com · +91 96064 85937

EMPOWERING FUTURE

COMPANY OVERVIEW

About Us

Mecwin Group is a Bengaluru-based engineering and manufacturing company specializing in motors, controllers, powertrains, wiring harnesses, SCADA automation and IoT solutions. We develop **customized solutions for electric mobility, industrial automation, agriculture, renewable energy, aerospace, defence and other demanding applications.**Our 50,000 sq. ft. manufacturing facility is equipped with modern machinery, engineering laboratories and advanced testing facilities. With strong in-house engineering capabilities, we manage the complete product journey — from design and prototyping to testing, validation and production.We work closely with OEMs to provide high-performance, reliable and application-specific solutions.

Our growth is driven by innovation, engineering excellence, quality, integrity and long-term partnerships with our customers, suppliers and employees.

Vision

To become the preferred global technology partner for high-efficiency motors, controllers and powertrains—from agriculture to aviation—by delivering innovation, quality, reliability, on-time service and environmentally responsible manufacturing.

Mission

To become the preferred global technology partner for high-efficiency motors, controllers and powertrains—from agriculture to aviation—by delivering innovation, quality, reliability, on-time service and environmentally responsible manufacturing.

Core Values

Customer Success | Innovation | Excellence | Integrity | People & Partnerships | Environmental Responsibility.

Contents

StandardBLDCmotorand geared motor catalogue

Introduction

Model Designation 1

External Rotor BLDC Motors

DC Brushless External Rotor Motor 4014 Series 2
DC Brushless External Rotor Motor 5060 Series 3
DC Brushless External Rotor Motor 7050 Series 4

BLDC Motors and Geared Motors

DC Brushless Motor 28 mm Series 5
DC Brushless Planetary Gear Motor 28 mm Series 6
DC Brushless Planetary Gear Motor 28 Series (2838 Base Motor) 7
DC Brushless Motor 36 mm Series 8
DC Brushless Planetary Gear Motor 36 mm Series 9
DC Brushless Planetary Gear Motor 36 Series (3650 Base Motor) 10
DC Brushless Motor 42 mm Series (4-Pole) 11
DC Brushless Motor 42 mm Series (8-Pole) 12
DC Brushless Planetary Gear Motor 42 Series (4260 Base Motor) 13
DC Brushless Planetary Gear Motor 42 mm Series (24 V Base Motor) 14
DC Brushless Planetary Gear Motor 42 mm Series (48 V Base Motor) 15
DC Brushless Motor 57 mm Series 16
DC Brushless Planetary Gear Motor 57 mm Series 17
DC Brushless RV25 Worm Gear Motor 57 mm Series 18
DC Brushless RV30 Worm Gear Motor 57 mm Series 19
DC Brushless Motor 60 mm Series 20
DC Brushless Planetary Gear Motor 60 mm Series 21
DC Brushless RV25 Worm Gear Motor 60 mm Series 22
DC Brushless RV30 Worm Gear Motor 60 mm Series 23
DC Brushless Motor 70 mm Series 24
DC Brushless Motor 80 mm Series (4-Pole) 25
DC Brushless Motor 80 mm Series (8-Pole) 26
DC Brushless Planetary Gear Motor 80 mm Series 27

Contents (continued)

DC Brushless Motor 86 mm Series 28
DC Brushless Planetary Gear Motor 86 mm Series 29
DC Brushless RV30 Worm Gear Motor 86 mm Series 30
DC Brushless RV40 Worm Gear Motor 86 mm Series 31
DC Brushless RV50 Worm Gear Motor 86 mm Series 32
DC Brushless Motor 90 mm Series 33
DC Brushless Motor 103 mm Series 34
DC Brushless Motor 110 mm Series 35
DC Brushless Planetary Gear Motor 110 mm Series 36
DC Brushless RV50 Worm Gear Motor 110 mm Series 37
DC Brushless RV63 Worm Gear Motor 110 mm Series 38
DC Brushless Motor QY71 Series 39
DC Brushless Motor QY90 Series 40
DC Brushless Motor QY100 Series 41
DC Brushless Motor QY112 Series 42
DC Brushless Motor QY132 Series 43

Product Technical Information

Motor Selection Method & Technical Terms 44
Technical Parameter Definitions 45
Unit Conversion Table & Motor Temperature Rise 46
Insulation Classes & EMC Noise Suppression 47
Motor Control Methods 48
PWM Control 49
Motor and Gearbox Combinations 50
Troubleshooting - DC Brake 51
Spur Gearbox & Gear Efficiency 52
Troubleshooting - Gear Reducer 53

Model Designation

HowtoreadaMecwinbrushlessDCmotor and geared motor part number

MEC	-	57	BLDC	-	24	30S	-	RV30	-	20K	-	M
1		2	3		4	5		6		7		8

Example: MEC-57BLDC-2430S-RV30-20K-M- 57 mm BLDC motor, 24 V DC, 3000 rpm, RV30 worm gearbox, 1:20 ratio, with brake

Part Number Structure

1	**Company Name**	MECWIN - Mecwin Technologies India Pvt. Ltd.
2	**Motor Frame Size**	28 mm, 36 mm, 42 mm, 57 mm, 60 mm, 70 mm, 80 mm, 86 mm, 90 mm, 103 mm, 110 mm External rotor: 4014, 5060, 7050 | IEC-style housing: QY71, QY90, QY100, QY112, QY132
3	**Motor Type**	BLDC - Brushless DC motor
4	**Supply Voltage**	Nominal DC supply voltage. Example: 24 = 24 V DC
5	**Motor Speed**	Rated speed in hundreds of rpm. Example: 30S = 3000 rpm
6	**Gearbox Type and Model**	RV - RV worm gear reducer (RV25, RV30, RV40, RV50, RV63) PG - Planetary gearbox (PG28, PG36, PG42, PG56, PG80, PG86, PG110)
7	**Reduction Ratio**	K = ratio. Example: 20K = 1:20 reduction
8	**Optional Accessories**	M - Electromagnetic brake BFE - Closed-loop feedback, i.e. motor fitted with an encoder (including simple hall encoder)

Standard Options

Encoders & Feedback
Hall-effect feedback as standard; incremental or absolute encoders available on request (BFE).

Brakes
Spring-applied electromagnetic brakes for holding and emergency stop duty (M).

Gearboxes
Planetary (PG) and RV worm (RV) reducers factory-fitted and tested with the motor.

Drives & Controllers
Matching Mecwin BLDC drives, speed potentiometers and IoT-enabled controllers.

Product Range at a Glance

	Frame Sizes	Output Range	Supply Voltage	Catalogue Pages
Round frame BLDC motors	4014, 5060, 7050	240 - 1120 W	18 - 48 V DC	2 - 4
Round frame BLDC motors	28, 36, 42, 80, 103 mm	3.7 - 1000 W	12 - 310 V DC	5, 8, 11, 26, 34
Square frame BLDC motors	42, 57, 60, 70, 80, 86, 90, 110 mm	26 W - 3.0 kW	24 - 310 V DC	12, 16, 20, 24, 25, 28, 33, 35
Housed BLDC motors (QY seriesQ)	Y71, QY90, QY100, QY112, QY132	0.5 - 15 kW	24 - 120 V DC	39 - 43
Planetary geared BLDC motors	PG28 to PG110	Rated torque up to 160 N·mas base motor		6, 7, 9, 10, 13 - 15, 17, 21, 27, 29, 36
Worm geared BLDC motors	RV25 to RV63	Rated torque up to 166 N·mas base motor		18, 19, 22, 23, 30 - 32, 37, 38

DC Brushless External Rotor Motor 4014 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**24**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 3000 h**

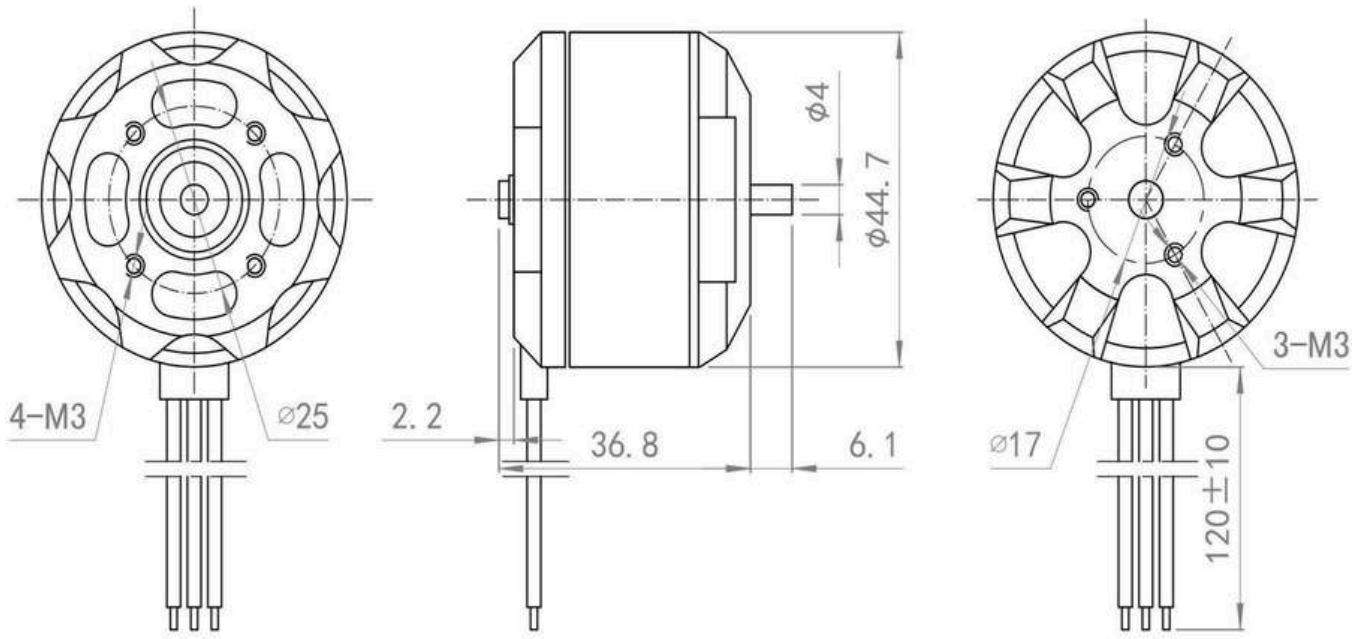
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class and matching drives are also available.

Motor Technical Data

Model	Voltage	KV	No-load		Rated Load				Stall	
			Speed	Current	Power	Speed	Torque	Current	Torque	Current
	V	rpm/V	rpm	A	W	rpm	N·m	A	N·m	A
MEC-4014BLDC-1852S	18	340	6120	1.1	240	5202	0.3	12.0	0.80	25
MEC-4014BLDC-2264S	22.2	340	7548	1.1	280	6416	0.4	12.0	0.90	25
MEC-4014BLDC-1861S	18	400	7200	1.3	300	6120	0.4	15.0	1.00	30
MEC-4014BLDC-2275S	22.2	400	8880	1.3	340	7548	0.5	15.0	1.10	30

Mechanical Dimensions



All dimensions in mm. Drawings are not to scale.

DC Brushless External Rotor Motor 5060 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**14**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 3000 h**

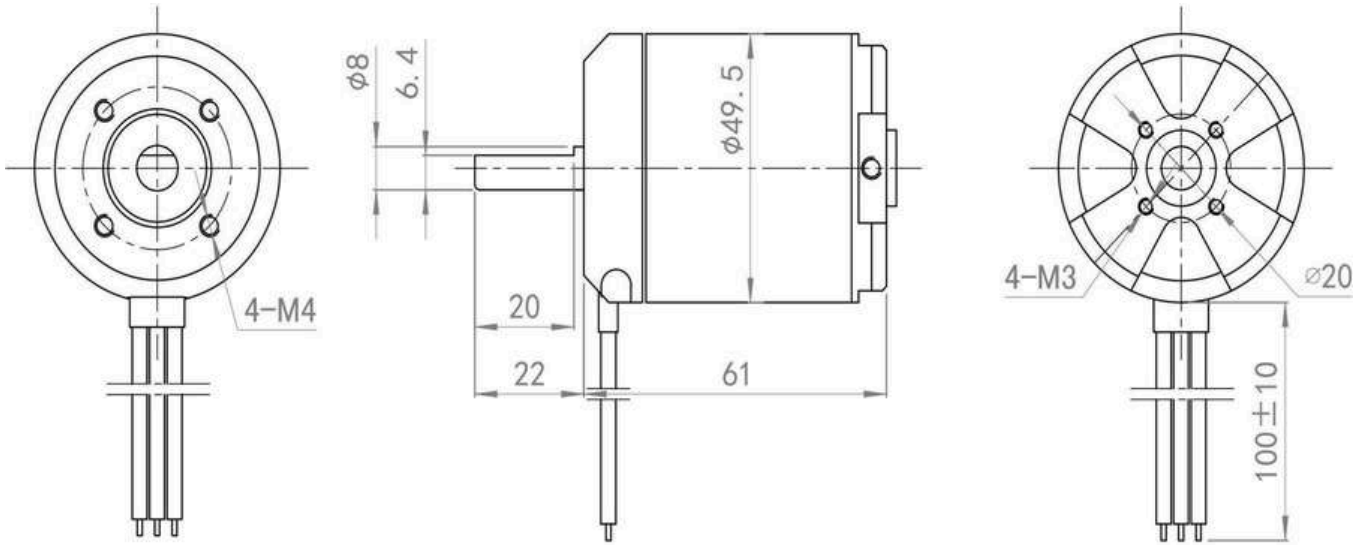
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class and matching drives are also available.

Motor Technical Data

Model	Voltage	KV	No-load		Rated Load				Stall	
			Speed	Current	Power	Speed	Torque	Current	Torque	Current
	V	rpm/V	rpm	A	W	rpm	N·m	A	N·m	A
MEC-5060BLDC-2480S	24	200	4800	0.8	450	4080	0.6	18.0	1.40	52
MEC-5060BLDC-3661S	36	200	7200	0.9	650	6120	0.8	18.0	2.10	52
MEC-5060BLDC-4881S	48	200	9600	1.1	850	8160	0.9	18.0	2.40	52
MEC-5060BLDC-3685S	36	280	10080	1.6	650	8568	0.7	18.0	1.90	52

Mechanical Dimensions



All dimensions in mm. Drawings are not to scale.

DC Brushless External Rotor Motor 7050 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**22**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 3000 h**

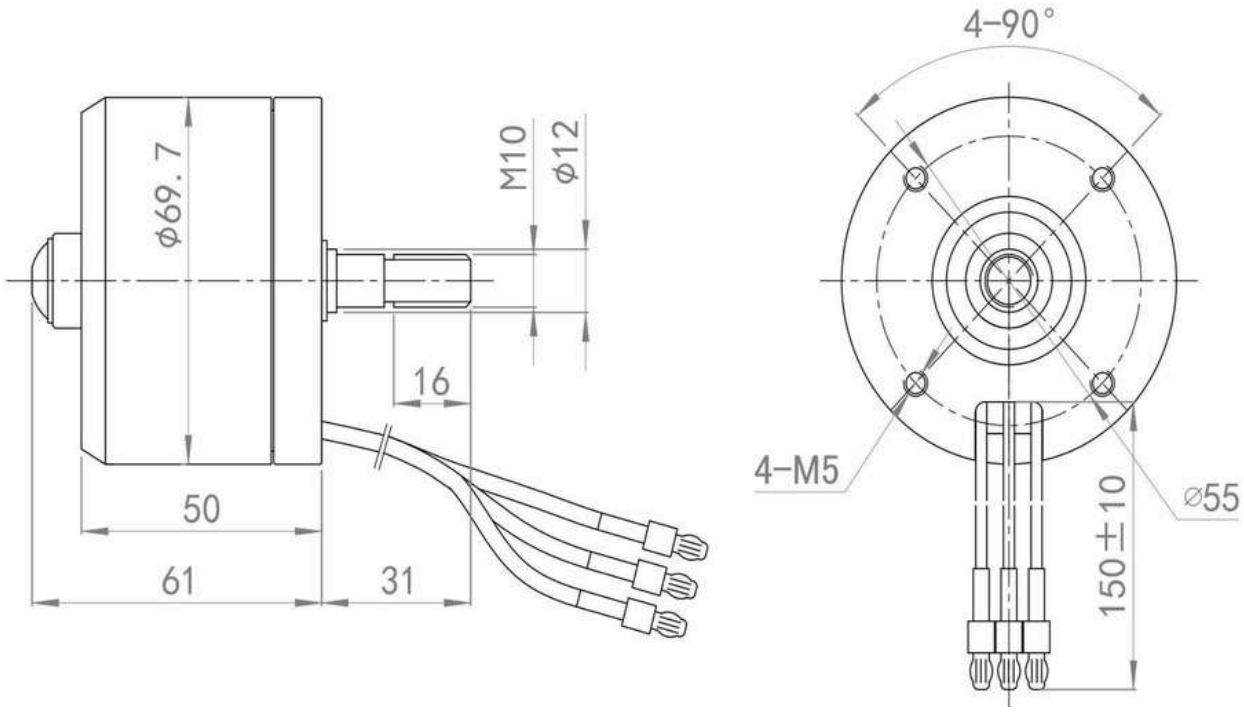
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class and matching drives are also available.

Motor Technical Data

Model	Voltage	KV	No-load		Rated Load				Stall	
			Speed	Current	Power	Speed	Torque	Current	Torque	Current
	V	rpm/V	rpm	A	W	rpm	N·m	A	N·m	A
MEC-7050BLDC-3627S	36	90	3240	0.9	980	2754	3.2	27.0	6.50	78
MEC-7050BLDC-2440S	24	200	4800	1.9	700	4080	0.9	29.0	2.50	78
MEC-7050BLDC-3661S	36	200	7200	2.4	980	6120	1.3	27.0	3.70	78
MEC-7050BLDC-4881S	48	200	9600	2.8	1120	8160	1.5	23.0	4.20	68

Mechanical Dimensions



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor 28 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**4**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 3000 h**

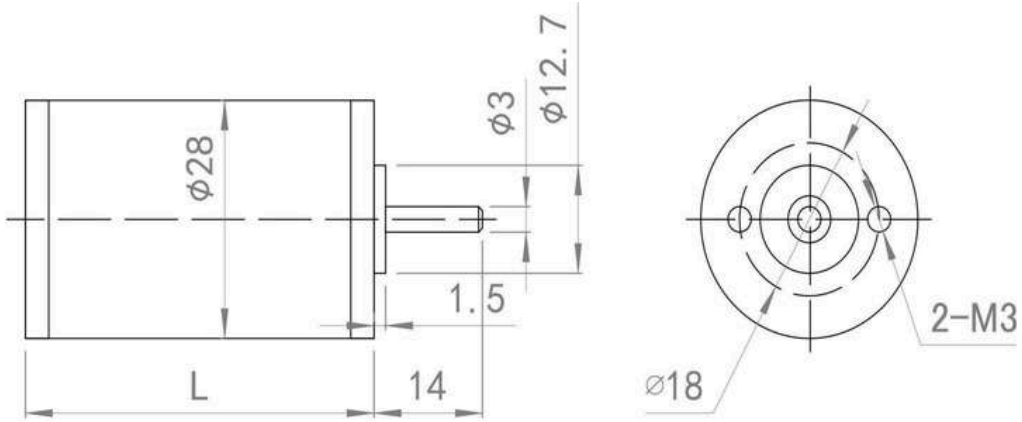
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-28BLDC-1230S	12	4000	0.20	3.7	3000	0.012	0.4	0.18	1.23	37.8
MEC-28BLDC-1230S-1	12	4000	0.2	4.7	3000	0.015	0.5	0.38	1.57	40.5
MEC-28BLDC-2430S	24	4000	0.60	6.3	3000	0.020	0.4	0.55	1.05	53
MEC-28BLDC-2430S-1	24	4000	0.4	15	3000	0.048	0.8	0.75	2.50	78

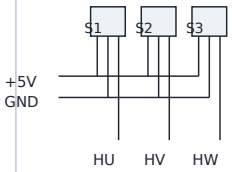
Mechanical Dimensions



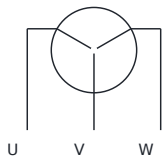
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



www.mecwinindia.com

DC Brushless Planetary Gear Motor 28 mm Series



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 30 N**
Max. Axial Load	**≤ 10 N**
Axial Play of Shaft	**≤ 0.03 mm**
Rated Permissible Torque	**2 N·m**

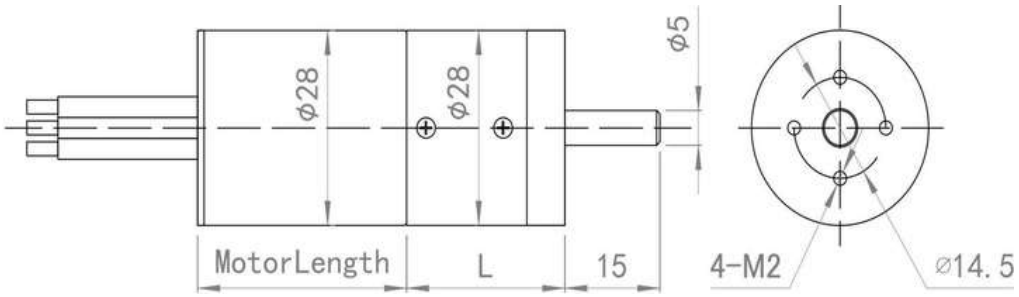
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-28BLDC-2430S	24	4000	0.60	6.3	3000	0.020	0.4	0.55	1.05	53
MEC-28BLDC-2430S-1	24	4000	0.4	15	3000	0.048	0.8	0.75	2.50	78

Gear Motor Technical Data

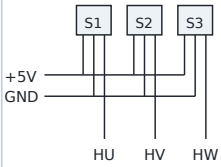
Model	Reduction ratio	1:X	1:4	1:14	1:19	1:27	1:51	1:71	1:100	1:139
MEC-28BLDC-2430S-1-PG28	Gearbox length	mm	13.9	32.4			43.9			
	No-load speed	rpm	1000	286	211	148	78	56	40	29
	Rated speed	rpm	750	214	158	111	59	42	30	22
	Rated torque	N·m	0.17	0.54	0.73	1.04	1.83	2.00	2.00	2.00

Mechanical Dimensions

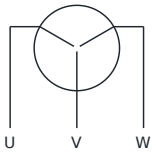


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Planetary Gear Motor 28 Series (2838 Base Motor)



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 30 N**
Max. Axial Load	**≤ 10 N**
Axial Play of Shaft	**≤ 0.03 mm**
Rated Permissible Torque	**2 N·m**

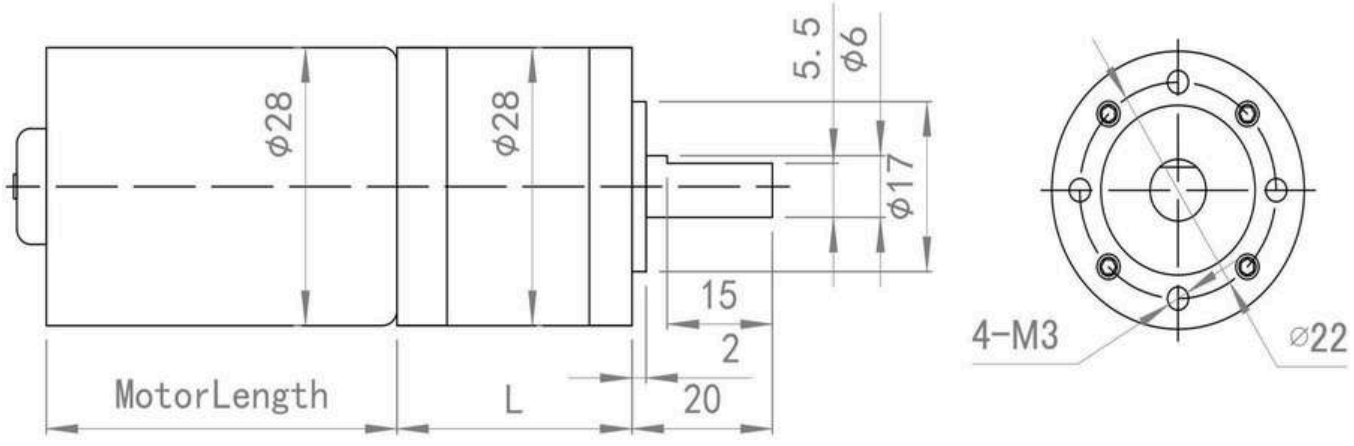
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-2838BLDC-1242S	12	5000	0.14	4.3	4200	0.01	0.6	0.03	1.65	37.8
MEC-2838BLDC-2444S	24	5000	0.09	4.5	4400	0.01	0.3	0.03	0.90	37.8

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:4	1:14	1:19	1:27	1:51	1:71	1:100	1:139
MEC-2838BLDC-1242S-PG28	Gearbox length	mm	28.5	36.0			43.0			
	No-load speed	rpm	1250	357	263	185	98	70	50	36
	Rated speed	rpm	1050	300	221	156	82	59	42	30
	Rated torque	N·m	0.04	0.11	0.15	0.22	0.38	0.53	0.75	1.04

Mechanical Dimensions



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor 36 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**4**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 3000 h**

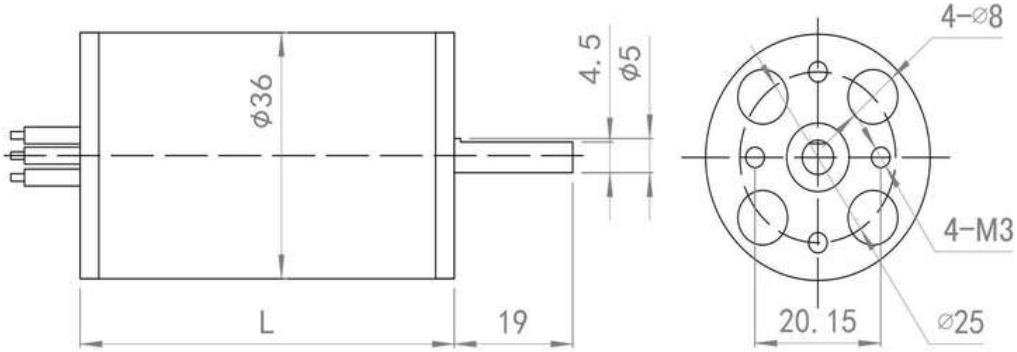
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-36BLDC-1218S	12	2400	0.10	7	1835	0.036	0.8	0.10	2.70	60
MEC-36BLDC-1225S	12	3200	0.1	9.4	2500	0.036	1.0	0.1	2.70	60
MEC-36BLDC-1830S	18	4000	0.10	11	3000	0.036	0.8	0.12	2.82	60
MEC-36BLDC-2440S	24	5200	0.1	15	4000	0.035	0.8	0.75	2.49	60

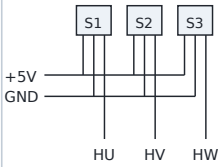
Mechanical Dimensions



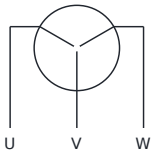
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless Planetary Gear Motor 36 mm Series



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 50 N**
Max. Axial Load	**≤ 20 N**
Axial Play of Shaft	**≤ 0.03 mm**
Rated Permissible Torque	**3 N·m**

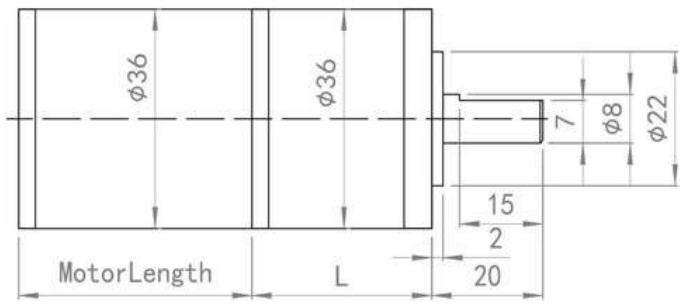
Motor Technical Data

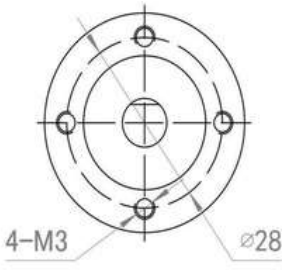
Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-36BLDC-1830S	18	4000	0.10	11	3000	0.036	0.8	0.12	2.82	60
MEC-36BLDC-2440S	24	5200	0.1	15	4000	0.035	0.8	0.75	2.49	60

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:4	1:14	1:19	1:27	1:51	1:71	1:100	1:139
MEC-36BLDC-1830S-PG36	Gearbox length	mm	24.8	32.4			41.9			
	No-load speed	rpm	1000	286	211	148	78	56	40	29
	Rated speed	rpm	750	214	158	111	59	42	30	22
	Rated torque	N·m	0.13	0.41	0.55	0.79	1.38	1.92	2.70	3.00

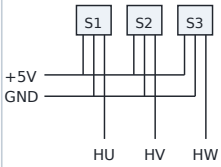
Mechanical Dimensions



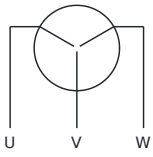


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Planetary Gear Motor 36 Series (3650 Base Motor)



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 50 N**
Max. Axial Load	**≤ 20 N**
Axial Play of Shaft	**≤ 0.03 mm**
Rated Permissible Torque	**3 N·m**

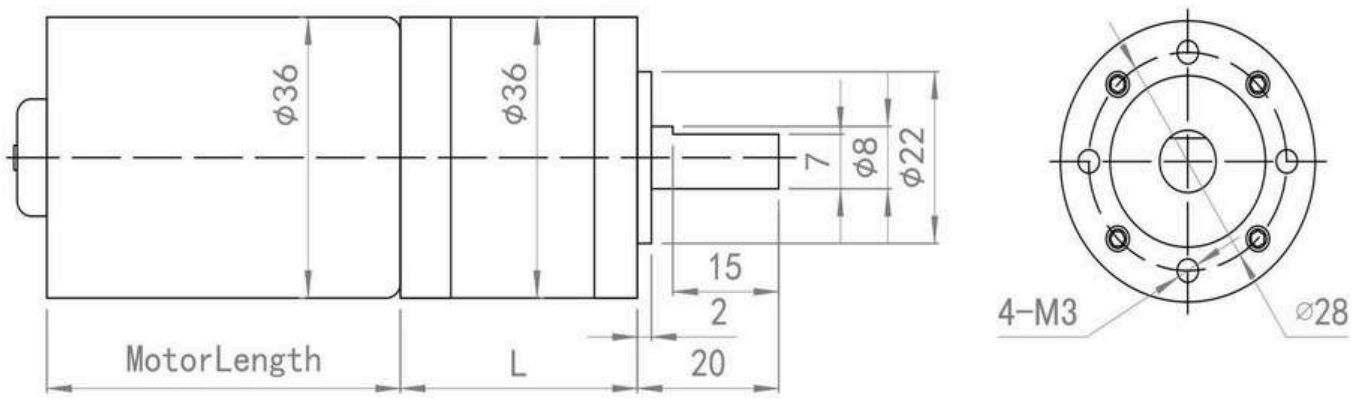
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-3650BLDC-1253S	12	6500	0.25	11	5300	0.02	1.4	0.06	4.20	50
MEC-3650BLDC-2434S	24	4000	0.08	7	3400	0.02	0.4	0.06	1.26	50

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:4	1:14	1:19	1:27	1:51	1:71	1:100	1:139
MEC-3650BLDC-1253S-PG36	Gearbox length	mm	26.0	33.5			40.5			
	No-load speed	rpm	1625	464	342	241	127	92	65	47
	Rated speed	rpm	1325	379	279	196	104	75	53	38
	Rated torque	N·m	0.07	0.22	0.31	0.43	0.76	1.06	1.49	2.07

Mechanical Dimensions



All dimensions in mm. Drawings are not to scale.

DC Brushless Motor 42 mm Series (4-Pole)



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**4**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 3000 h**

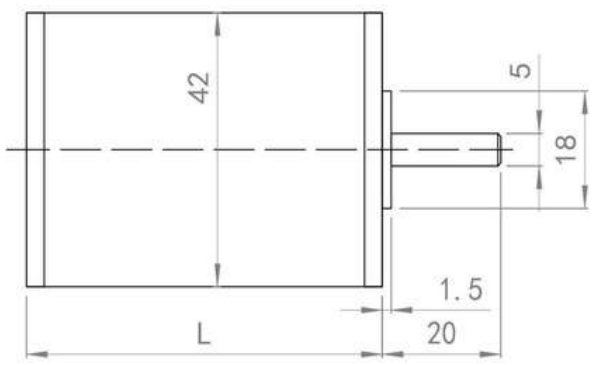
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-42BLDC-1230S	12	4000	0.10	11	3000	0.035	0.6	0.18	1.89	42
MEC-42BLDC-1230S-1	12	4000	0.13	17	3000	0.054	1.2	0.38	3.60	60
MEC-42BLDC-2430S	24	4000	0.20	25	3000	0.080	1.8	0.55	5.31	60
MEC-42BLDC-2430S-1	24	4000	0.3	41	3000	0.130	2.5	0.75	7.59	78

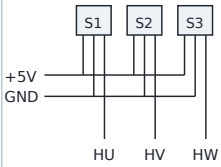
Mechanical Dimensions



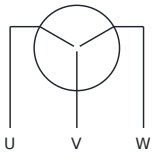


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor 42 mm Series (8-Pole)



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

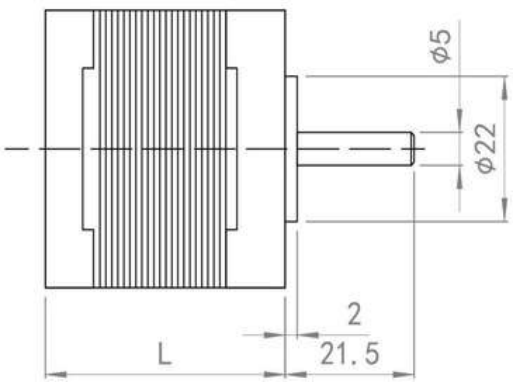
Customisable Solutions

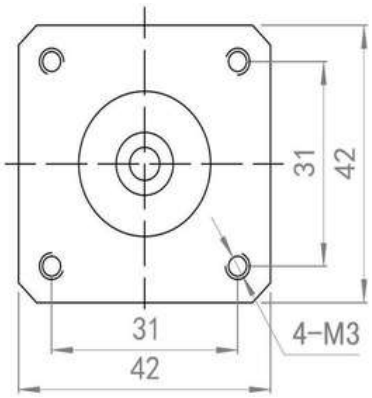
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-42BLDC-2440S	24	5200	0.20	26	4000	0.063	1.8	0.18	5.40	40
MEC-42BLDC-4840S	48	5200	0.2	52	4000	0.125	1.8	0.38	5.25	60
MEC-42BLDC-2440S-1	24	5200	0.60	78	4000	0.185	5.2	0.55	15.60	80
MEC-42BLDC-4840S-1	48	5200	0.4	105	4000	0.250	3.3	0.75	9.90	100

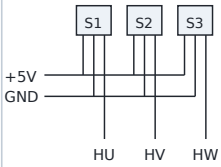
Mechanical Dimensions



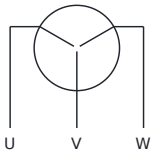


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Planetary Gear Motor 42 Series (4260 Base Motor)



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 80 N**
Max. Axial Load	**≤ 30 N**
Axial Play of Shaft	**≤ 0.03 mm**
Rated Permissible Torque	**6 N·m**

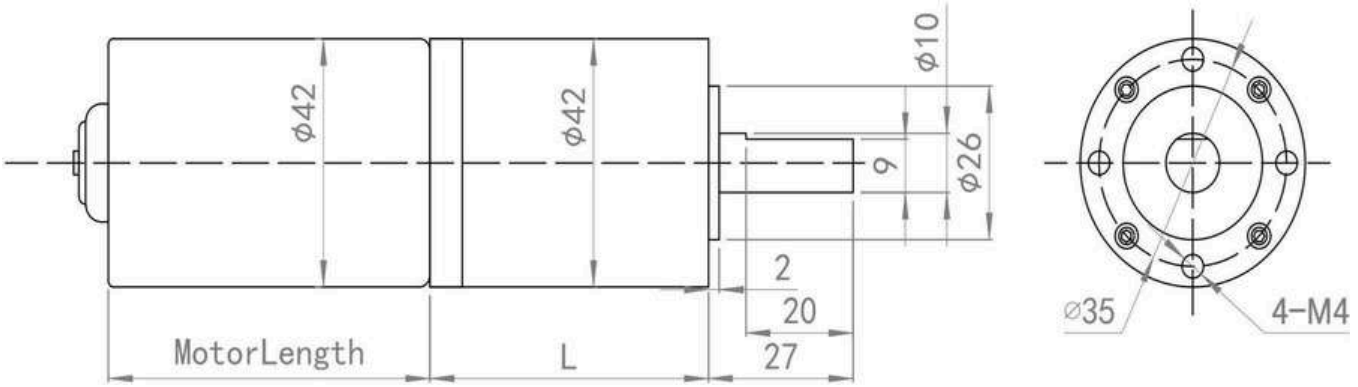
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-4260BLDC-1234S	12	4533	0.22	17	3400	0.05	1.8	0.14	5.31	60
MEC-4260BLDC-2435S	24	4733	0.12	18	3550	0.05	0.9	0.15	2.81	60

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:4	1:14	1:19	1:27	1:51	1:71	1:100	1:139
MEC-4260BLDC-1234S-PG42	Gearbox length	mm	31.5	42.0			52.0			
	No-load speed	rpm	1133	324	239	168	89	64	45	33
	Rated speed	rpm	850	243	179	126	67	48	34	24
	Rated torque	N·m	0.17	0.54	0.73	1.04	1.83	2.54	3.58	4.98

Mechanical Dimensions



All dimensions in mm. Drawings are not to scale.

DC Brushless Planetary Gear Motor 42 mm Series (24 V Base Motor)



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 80 N**
Max. Axial Load	**≤ 30 N**
Axial Play of Shaft	**≤ 0.03 mm**
Rated Permissible Torque	**9 N·m**

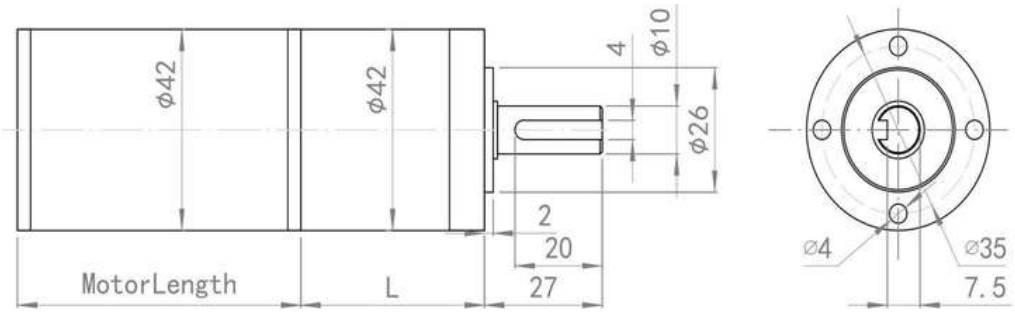
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-42BLDC-2430S	24	4000	0.20	25	3000	0.080	1.8	0.55	5.31	60
MEC-42BLDC-2430S-1	24	4000	0.3	41	3000	0.130	2.5	0.75	7.59	78

Gear Motor Technical Data

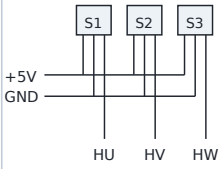
Model	Reduction ratio	1:X	1:4	1:15	1:18	1:36	1:54	1:81	1:126	1:216
MEC-42BLDC-2430S-1-PG42	Gearbox length	mm	32.5	46.3			60.1			
	No-load speed	rpm	1000	267	222	111	74	49	32	19
	Rated speed	rpm	750	200	167	83	56	37	24	14
	Rated torque	N·m	0.47	1.58	1.90	3.79	5.27	7.90	9.00	9.00

Mechanical Dimensions

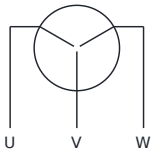


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Planetary Gear Motor 42 mm Series (48 V Base Motor)



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 50 N**
Max. Axial Load	**≤ 30 N**
Axial Play of Shaft	**≤ 0.03 mm**
Rated Permissible Torque	**9 N·m**

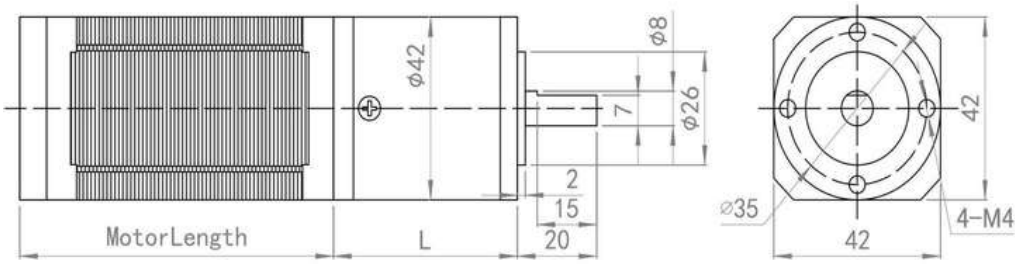
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-42BLDC-2440S-1	24	5200	0.60	78	4000	0.185	5.2	0.55	15.60	80
MEC-42BLDC-4840S-1	48	5200	0.4	105	4000	0.250	3.3	0.75	9.90	100

Gear Motor Technical Data

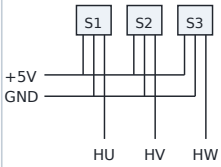
Model	Reduction ratio	1:X	1:4	1:14	1:19	1:27	1:51	1:71	1:100	1:139
MEC-42BLDC-4840S-1-PG42	Gearbox length	mm	26	34			41			
	No-load speed	rpm	1300	371	274	193	102	73	52	37
	Rated speed	rpm	1000	286	211	148	78	56	40	29
	Rated torque	N·m	0.90	2.84	3.85	5.47	9.00	9.00	9.00	9.00

Mechanical Dimensions

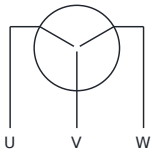


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor 57 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**4**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

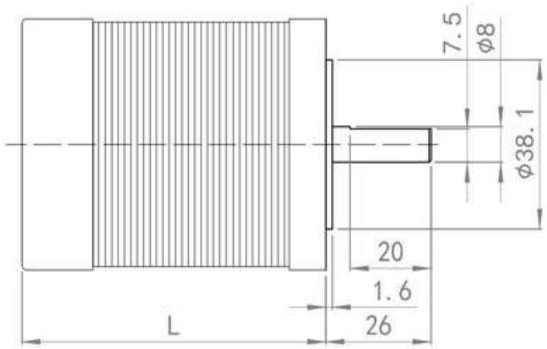
Customisable Solutions

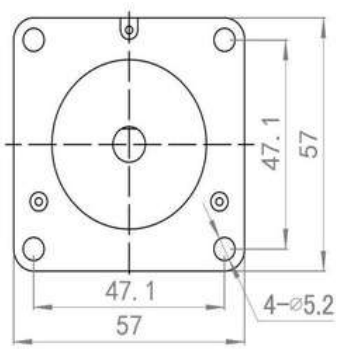
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-57BLDC-2430S	24	4000	0.41	63	3000	0.201	3.3	0.17	9.84	55
MEC-57BLDC-4830S	48	4000	0.41	125	3000	0.398	3.3	0.66	9.77	75
MEC-57BLDC-2430S-1	24	4000	1.22	188	3000	0.598	9.8	0.99	29.38	95
MEC-57BLDC-4830S-1	48	4000	0.65	200	3000	0.637	5.2	1.32	15.63	115

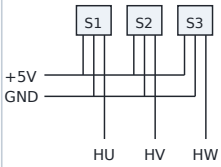
Mechanical Dimensions



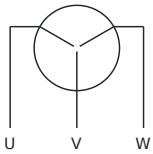


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Planetary Gear Motor 57 mm Series



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 250 N**
Max. Axial Load	**≤ 100 N**
Axial Play of Shaft	**≤ 0.08 mm**
Rated Permissible Torque	**30 N·m**

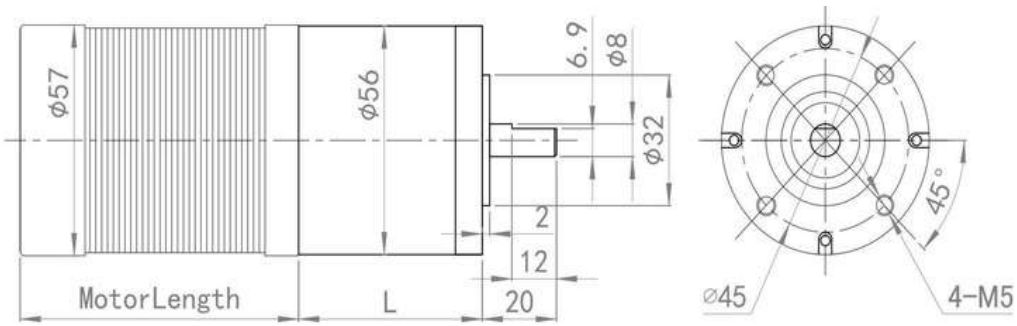
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-57BLDC-2430S-1	24	4000	1.22	188	3000	0.598	9.8	0.99	29.38	95
MEC-57BLDC-4830S-1	48	4000	0.65	200	3000	0.637	5.2	1.32	15.63	115

Gear Motor Technical Data

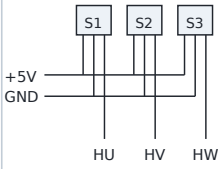
Model	Reduction ratio	1:X	1:3.6	1:13	1:15	1:23	1:47	1:65	1:96	1:121
MEC-57BLDC-4830S-1-PG56	Gearbox length	mm	37.8	49.5			60.8			
	No-load speed	rpm	1111	308	267	174	85	62	42	33
	Rated speed	rpm	833	231	200	130	64	46	31	25
	Rated torque	N·m	2.06	6.70	7.74	11.86	22.44	30.00	30.00	30.00

Mechanical Dimensions

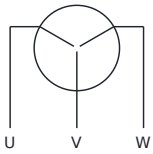


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless RV25 Worm Gear Motor 57 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**0.7 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**15 N·m**

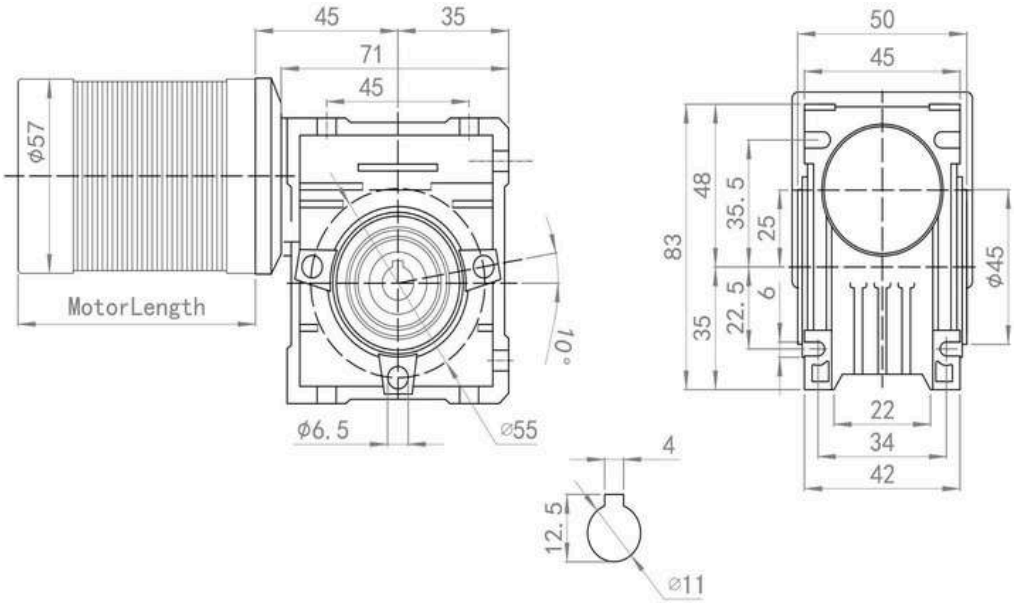
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-57BLDC-2430S-1	24	4000	1.22	188	3000	0.598	9.8	0.99	29.38	95
MEC-57BLDC-4830S-1	48	4000	0.65	200	3000	0.637	5.2	1.32	15.63	115

Gear Motor Technical Data

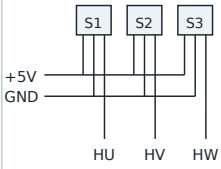
Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:25	1:30	1:40	1:50	1:60
MEC-57BLDC-4830S-1-RV25	No-load speed	rpm	533	267	200	160	133	100	80	67
	Rated speed	rpm	400	200	150	120	100	75	60	50
	Rated torque	N·m	2.39	4.78	6.37	7.96	9.55	12.73	15.00	15.00

Mechanical Dimensions

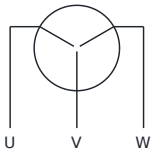


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless RV30 Worm Gear Motor 57 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**1.2 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**24 N·m**

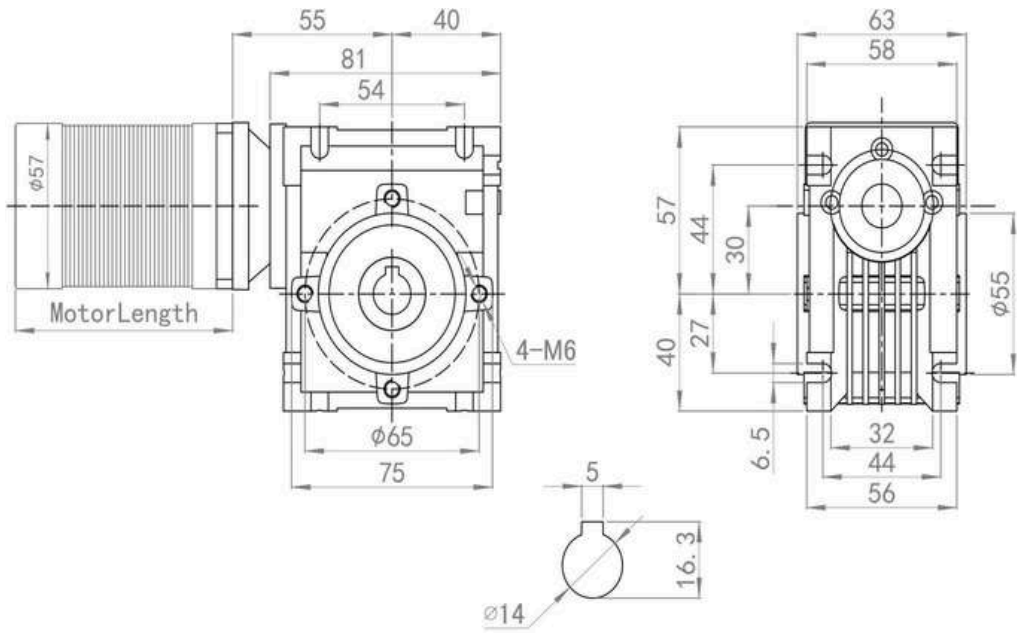
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-57BLDC-2430S-1	24	4000	1.22	188	3000	0.598	9.8	0.99	29.38	95
MEC-57BLDC-4830S-1	48	4000	0.65	200	3000	0.637	5.2	1.32	15.63	115

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:25	1:30	1:40	1:50	1:60
MEC-57BLDC-4830S-1-RV30	No-load speed	rpm	533	267	200	160	133	100	80	67
	Rated speed	rpm	400	200	150	120	100	75	60	50
	Rated torque	N·m	2.39	4.78	6.37	7.96	9.55	12.73	15.00	15.00

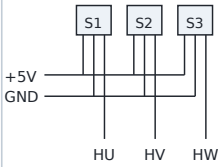
Mechanical Dimensions



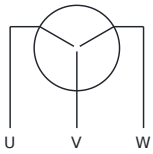
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless Motor 60 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

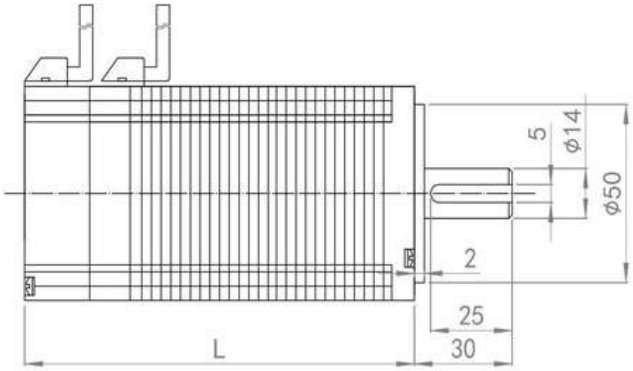
Customisable Solutions

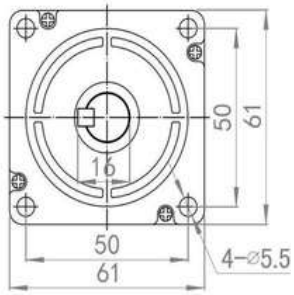
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-60BLDC-4830S	48	4000	0.38	100	3000	0.32	3.0	0.96	9.00	78
MEC-60BLDC-4830S-1	48	4000	0.75	200	3000	0.65	6.0	1.95	18.00	99
MEC-60BLDC-4830S-2	48	4000	1.10	300	3000	0.95	8.8	2.85	26.40	120
MEC-60BLDC-31030S	310	4000	0.22	400	3000	1.27	1.8	3.81	5.25	140

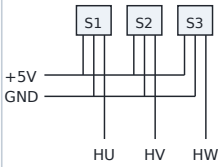
Mechanical Dimensions



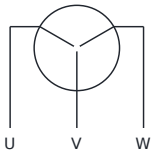


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Planetary Gear Motor 60 mm Series



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 250 N**
Max. Axial Load	**≤ 100 N**
Axial Play of Shaft	**≤ 0.08 mm**
Rated Permissible Torque	**45 N·m**

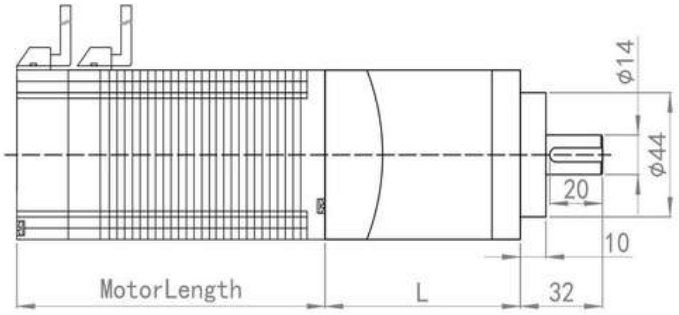
Motor Technical Data

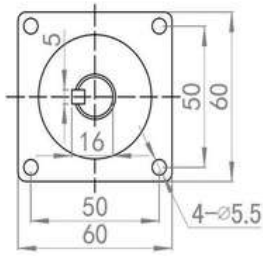
Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-60BLDC-4830S-2	48	4000	1.10	300	3000	0.95	8.8	2.85	26.40	120
MEC-60BLDC-31030S	310	4000	0.22	400	3000	1.27	1.8	3.81	5.25	140

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:4	1:10	1:24	1:36	1:64	1:96	1:144	1:216
MEC-60BLDC-31030S-PG80	Gearbox length	mm	74	88			102			
	No-load speed	rpm	1000	400	167	111	63	42	28	19
	Rated speed	rpm	750	300	125	83	47	31	21	14
	Rated torque	N·m	4.57	10.29	24.69	37.03	45.00	45.00	45.00	45.00

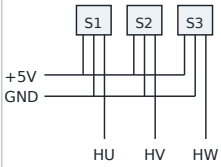
Mechanical Dimensions



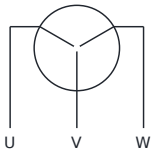


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless RV25 Worm Gear Motor 60 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**0.7 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**15 N·m**

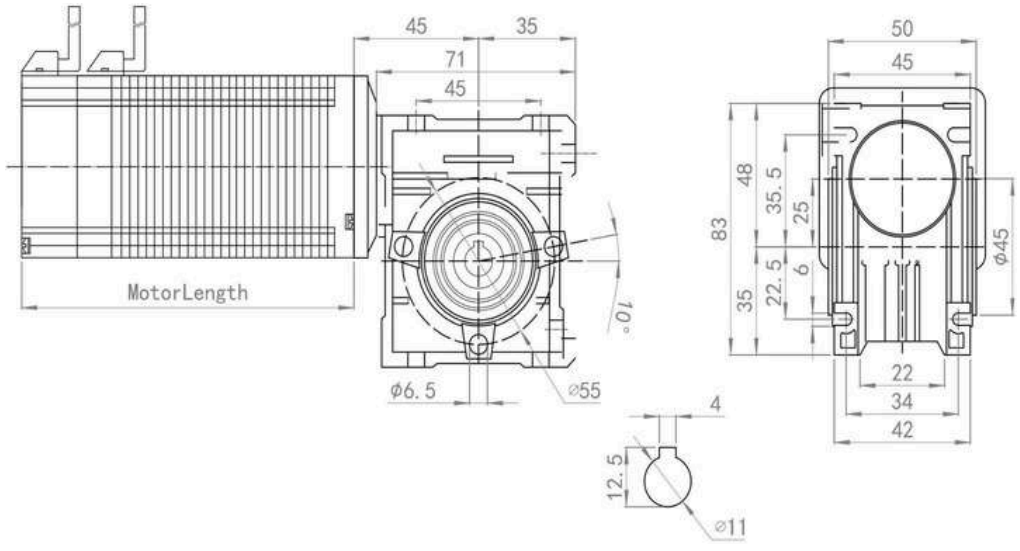
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-60BLDC-4830S-2	48	4000	1.10	300	3000	0.95	8.8	2.85	26.40	120
MEC-60BLDC-31030S	310	4000	0.22	400	3000	1.27	1.8	3.81	5.25	140

Gear Motor Technical Data

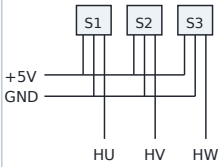
Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:25	1:30	1:40	1:50	1:60
MEC-60BLDC-31030S-RV25	No-load speed	rpm	533	267	200	160	133	100	80	67
	Rated speed	rpm	400	200	150	120	100	75	60	50
	Rated torque	N·m	4.76	9.53	12.70	15.00	15.00	15.00	15.00	15.00

Mechanical Dimensions

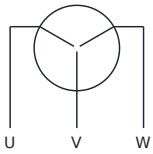


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless RV30 Worm Gear Motor 60 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**1.2 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**24 N·m**

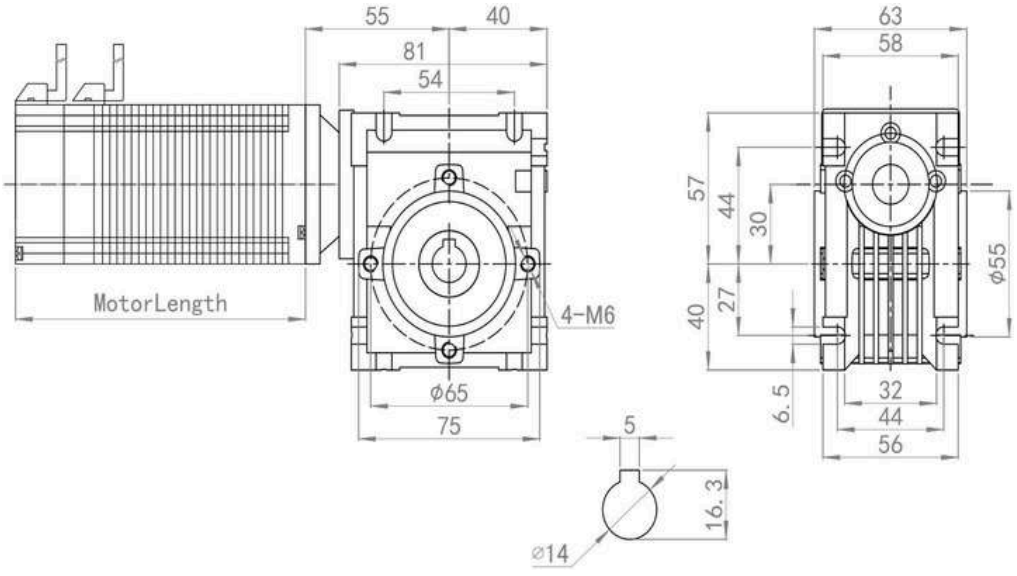
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-60BLDC-4830S-2	48	4000	1.10	300	3000	0.95	8.8	2.85	26.40	120
MEC-60BLDC-31030S	310	4000	0.22	400	3000	1.27	1.8	3.81	5.25	140

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:25	1:30	1:40	1:50	1:60
MEC-60BLDC-4830S-RV30	No-load speed	rpm	533	267	200	160	133	100	80	67
	Rated speed	rpm	400	200	150	120	100	75	60	50
	Rated torque	N·m	4.76	9.53	12.70	15.88	19.05	24.00	24.00	24.00

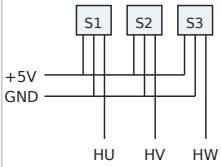
Mechanical Dimensions



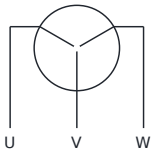
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



www.mecwinindia.com

DC Brushless Motor 70 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

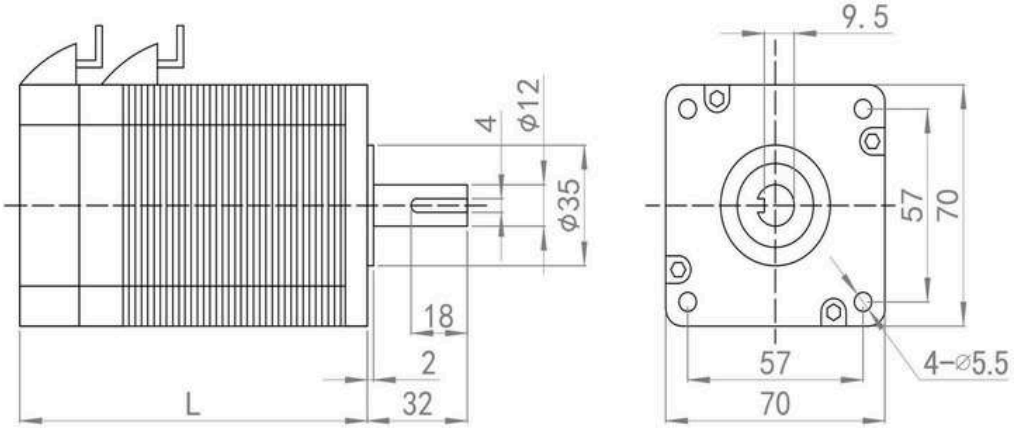
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-70BLDC-2430S	24	4000	1.09	157	3000	0.50	8.7	1.50	26.17	86
MEC-70BLDC-4830S	48	4000	0.55	157	3000	0.50	4.4	1.50	13.08	86
MEC-70BLDC-4830S-1	48	4000	1.09	314	3000	1.00	8.7	3.00	26.17	116
MEC-70BLDC-31030S	310	4000	0.25	471	3000	1.50	2.0	4.50	6.08	136

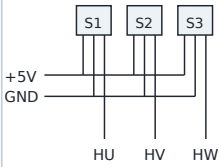
Mechanical Dimensions



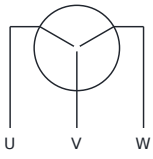
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



www.mecwinindia.com

DC Brushless Motor 80 mm Series (4-Pole)



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**4**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

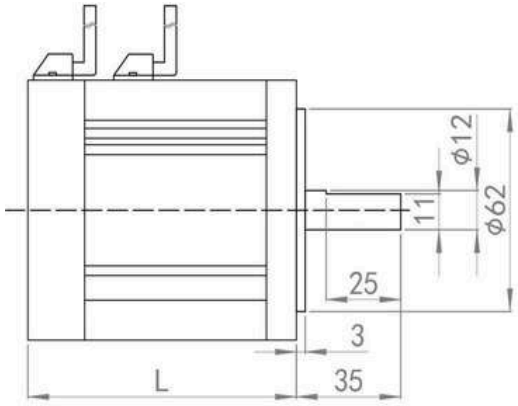
Customisable Solutions

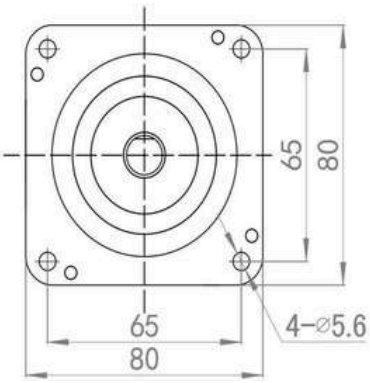
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-80BLDC-4830S	48	4000	0.41	126	3000	0.40	3.3	1.20	9.84	86
MEC-80BLDC-31060S	310	8000	0.22	440	6000	0.70	1.8	2.10	5.32	106
MEC-80BLDC-4830S-1	48	4000	1.30	400	3000	1.27	10.4	3.82	31.25	126
MEC-80BLDC-31030S	310	4000	0.38	750	3000	2.39	3.0	7.16	9.07	130

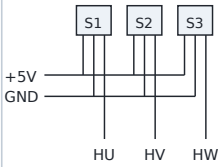
Mechanical Dimensions



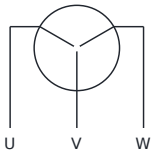


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Motor 80 mm Series (8-Pole)



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

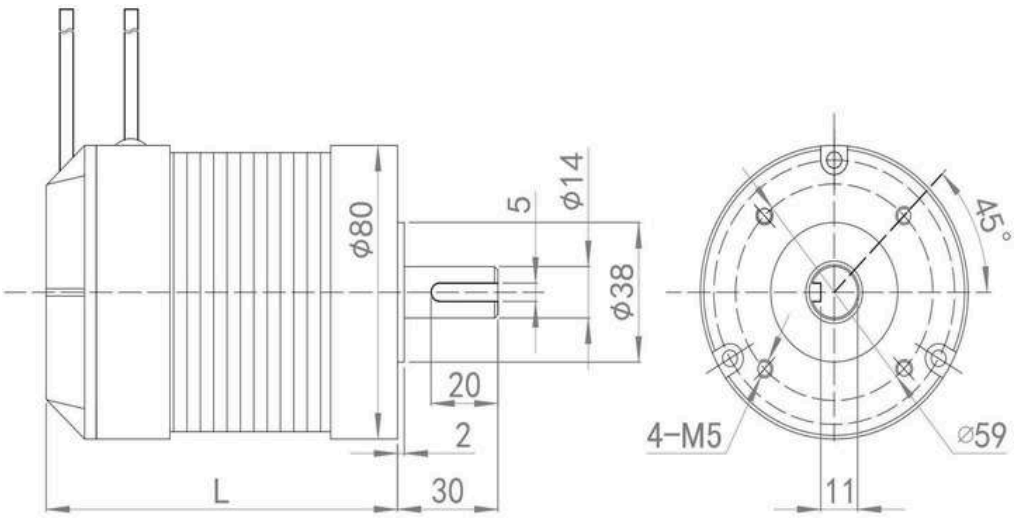
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-80BLDC-4830S	48	4000	0.41	126	3000	0.40	3.3	1.20	9.84	95
MEC-80BLDC-31060S	310	8000	0.11	220	6000	0.35	0.9	1.05	2.66	115
MEC-80BLDC-4830S-1	48	4000	1.30	400	3000	1.27	10.4	3.82	31.25	124
MEC-80BLDC-2460S	24	8000	1.95	300	6000	0.48	15.6	1.43	46.88	124

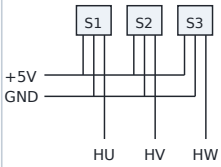
Mechanical Dimensions



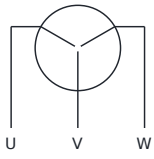
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



www.mecwinindia.com

DC Brushless Planetary Gear Motor 80 mm Series



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 500 N**
Max. Axial Load	**≤ 130 N**
Axial Play of Shaft	**≤ 0.1 mm**
Rated Permissible Torque	**60 N·m**

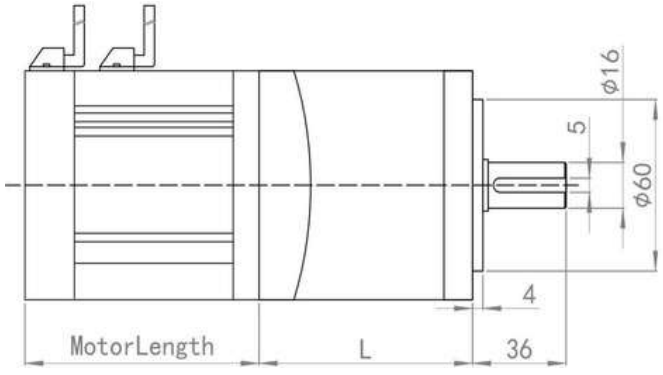
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-80BLDC-4830S-1	48	4000	1.30	400	3000	1.27	10.4	3.82	31.25	126
MEC-80BLDC-31030S	310	4000	0.38	750	3000	2.39	3.0	7.16	9.07	130

Gear Motor Technical Data

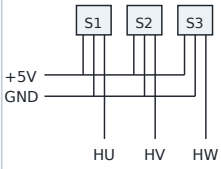
Model	Reduction ratio	1:X	1:4	1:10	1:24	1:36	1:64	1:96	1:144	1:216
MEC-80BLDC-31030S-PG80	Gearbox length	mm	82	96			110			
	No-load speed	rpm	1000	400	167	111	63	42	28	19
	Rated speed	rpm	750	300	125	83	47	31	21	14
	Rated torque	N·m	8.60	19.34	46.41	60.00	60.00	60.00	60.00	60.00

Mechanical Dimensions

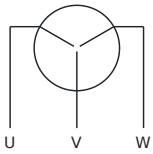


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Motor 86 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**4**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

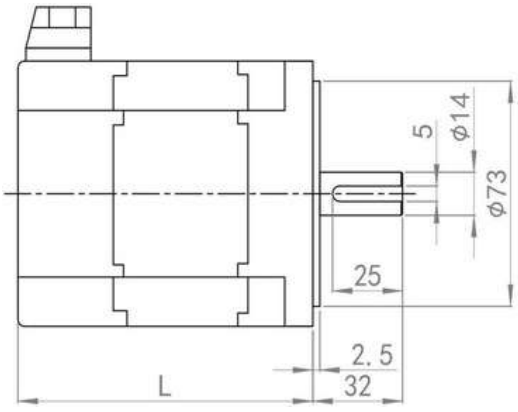
Customisable Solutions

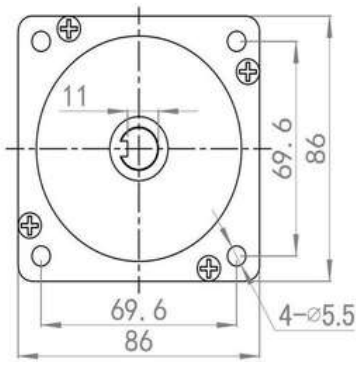
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-86BLDC-4830S	48	4000	1.02	314	3000	1.00	8.2	3.00	24.53	80
MEC-86BLDC-4830S-1	48	4000	1.30	400	3000	1.27	10.4	3.82	31.25	90
MEC-86BLDC-4830S-2	48	4000	2.15	659	3000	2.10	17.2	6.29	51.48	115
MEC-86BLDC-31030S	310	4000	0.40	785	3000	2.50	3.2	7.50	9.50	130

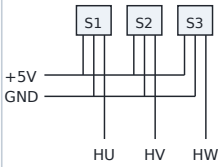
Mechanical Dimensions



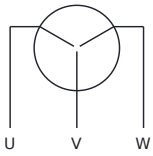


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Planetary Gear Motor 86 mm Series



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 500 N**
Max. Axial Load	**≤ 130 N**
Axial Play of Shaft	**≤ 0.1 mm**
Rated Permissible Torque	**80 N·m**

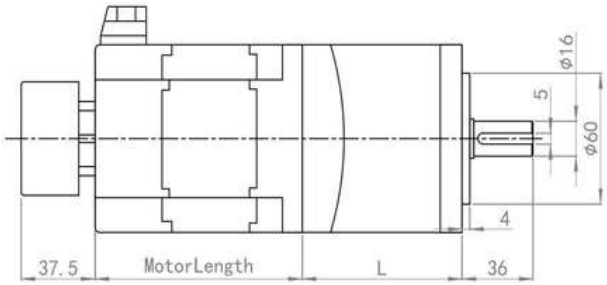
Motor Technical Data

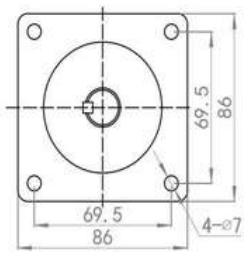
Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-86BLDC-4830S-2	48	4000	4.07	1250	3000	3.98	32.6	11.94	97.66	130
MEC-86BLDC-31030S	310	4000	0.40	785	3000	2.50	3.2	7.50	9.50	130

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:4	1:10	1:24	1:36	1:64	1:96	1:144	1:216
MEC-86BLDC-4830S-2-PG86	Gearbox length	mm	80	94			108			
	No-load speed	rpm	1000	400	167	111	63	42	28	19
	Rated speed	rpm	750	300	125	83	47	31	21	14
	Rated torque	N·m	14.33	32.23	77.36	80.00	80.00	80.00	80.00	80.00

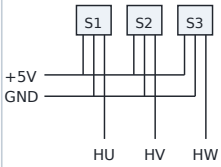
Mechanical Dimensions



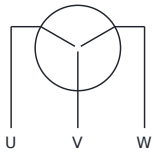


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless RV30 Worm Gear Motor 86 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**1.2 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**24 N·m**

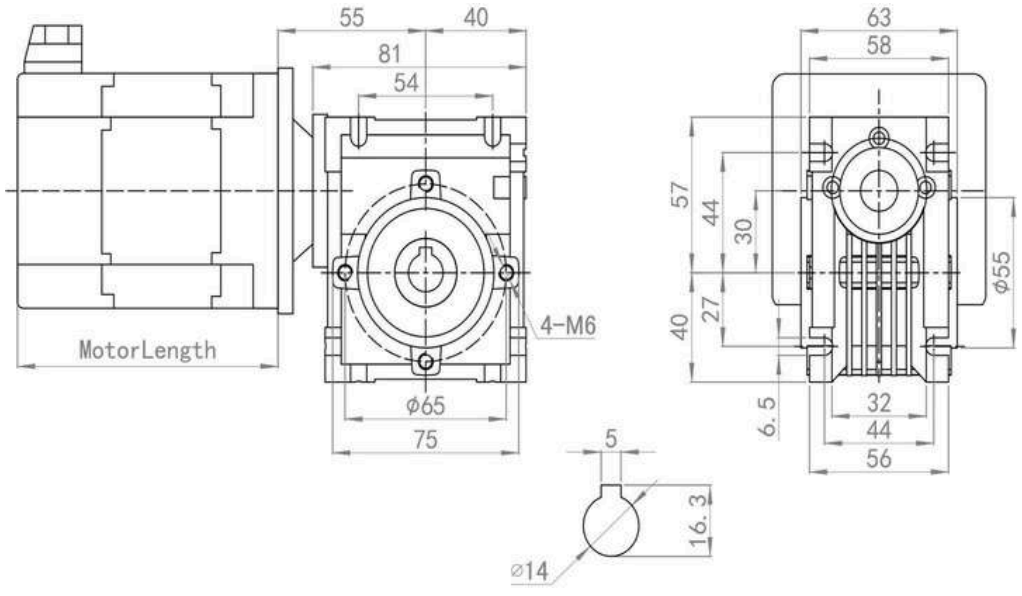
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-86BLDC-4830S-2	48	4000	2.15	659	3000	2.10	17.2	6.29	51.48	115
MEC-86BLDC-31030S	310	4000	0.40	785	3000	2.50	3.2	7.50	9.50	130

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:25	1:30	1:40	1:50	1:60
MEC-86BLDC-31030S-RV30	No-load speed	rpm	533	267	200	160	133	100	80	67
	Rated speed	rpm	400	200	150	120	100	75	60	50
	Rated torque	N·m	9.37	18.74	24.00	24.00	24.00	24.00	24.00	24.00

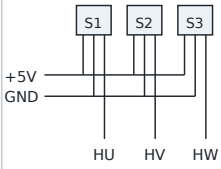
Mechanical Dimensions



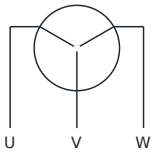
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless RV40 Worm Gear Motor 86 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**2.3 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**53 N·m**

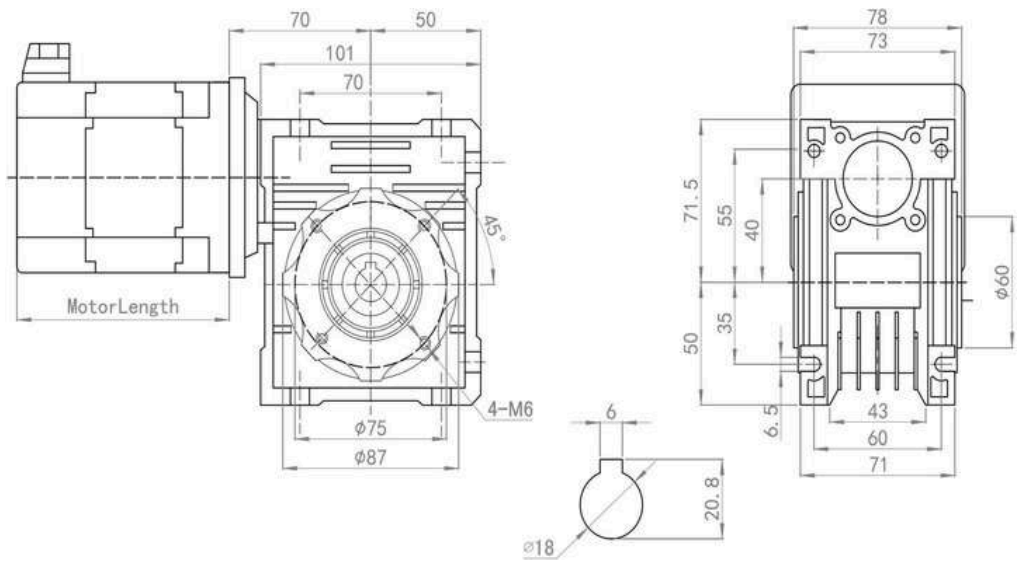
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-86BLDC-4830S-2	48	4000	2.15	659	3000	2.10	17.2	6.29	51.48	115
MEC-86BLDC-31030S	310	4000	0.40	785	3000	2.50	3.2	7.50	9.50	130

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:30	1:50	1:60	1:80	1:100
MEC-86BLDC-31030S-RV40	No-load speed	rpm	533	267	200	133	80	67	50	40
	Rated speed	rpm	400	200	150	100	60	50	38	30
	Rated torque	N·m	9.37	18.74	24.99	37.48	53.00	53.00	53.00	53.00

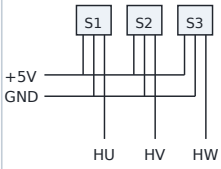
Mechanical Dimensions



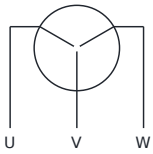
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless RV50 Worm Gear Motor 86 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**3.5 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**81 N·m**

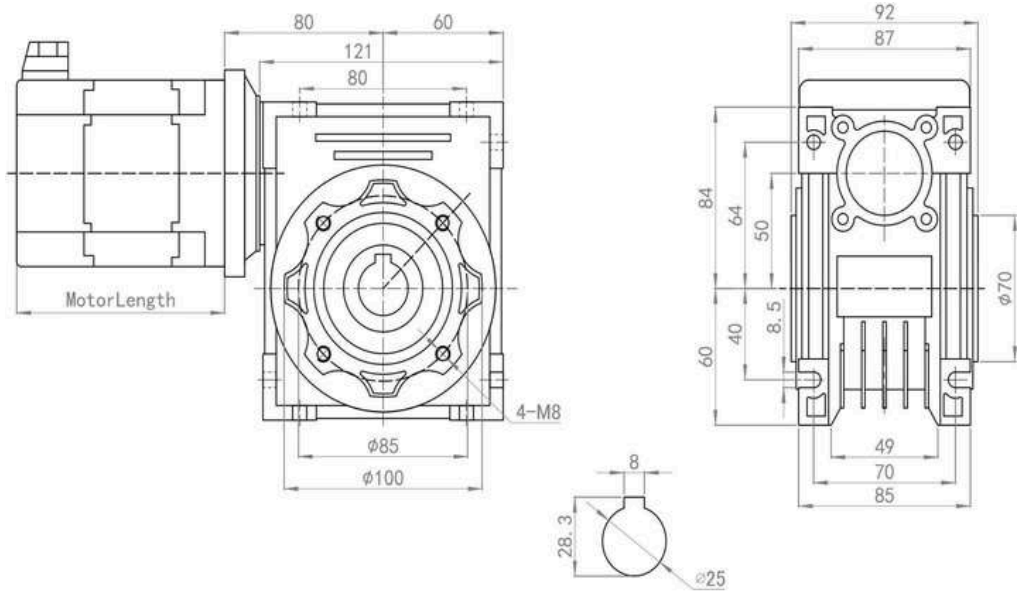
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-86BLDC-4830S-2	48	4000	2.15	659	3000	2.10	17.2	6.29	51.48	115
MEC-86BLDC-31030S	310	4000	0.40	785	3000	2.50	3.2	7.50	9.50	130

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:30	1:50	1:60	1:80	1:100
MEC-86BLDC-31030S-RV50	No-load speed	rpm	533	267	200	133	80	67	50	40
	Rated speed	rpm	400	200	150	100	60	50	38	30
	Rated torque	N·m	9.37	18.74	24.99	37.48	62.47	74.97	81.00	81.00

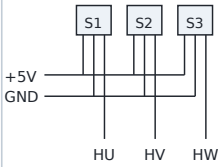
Mechanical Dimensions



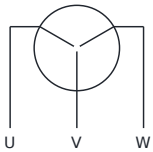
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless Motor 90 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

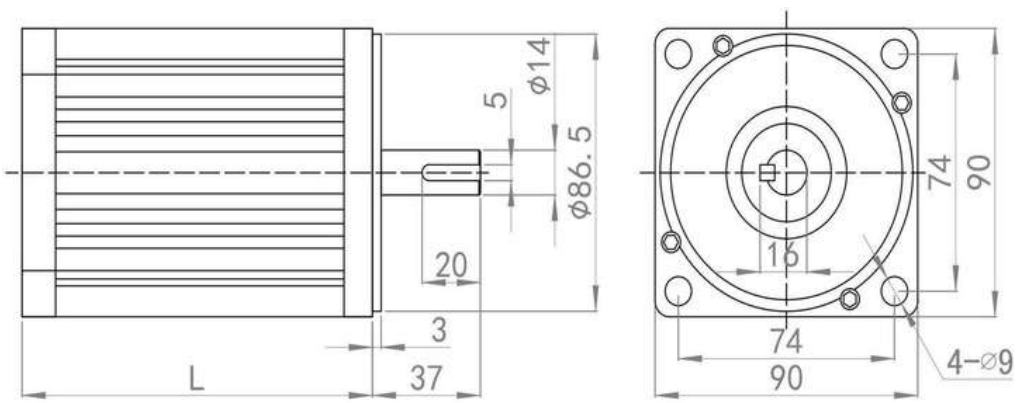
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-90BLDC-4830S	48	4000	1.02	314	3000	1.00	8.2	3.00	24.53	87
MEC-90BLDC-4830S-1	48	4000	1.84	565	3000	1.80	14.7	5.40	44.14	112
MEC-90BLDC-4830S-2	48	4000	2.15	659	3000	2.10	17.2	6.29	51.48	122
MEC-90BLDC-31030S	310	4000	0.40	785	3000	2.50	3.2	7.50	9.50	137

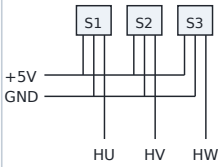
Mechanical Dimensions



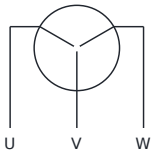
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless Motor 103 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material Design	**High-performance NdFeB**
Life	**≥ 5000 h**

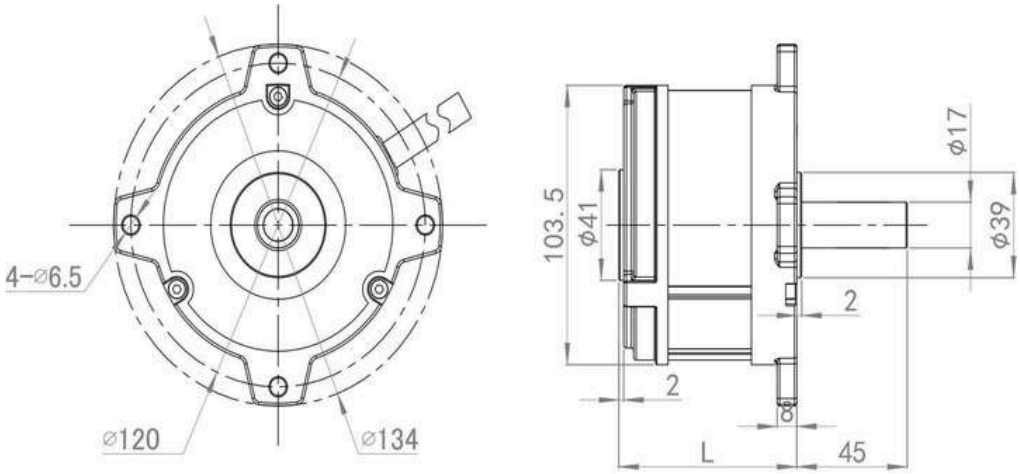
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-103BLDC-2430S	24	4000	2.28	350	3000	1.11	18.2	3.34	54.69	56.5
MEC-103BLDC-3627S	36	3600	2.39	550	2700	1.95	19.1	5.84	57.29	62
MEC-103BLDC-4828S	48	3700	2.44	750	2800	2.56	19.5	7.67	58.59	70
MEC-103BLDC-7230S	72	4700	2.17	1000	3500	2.73	17.4	8.19	52.08	72.5

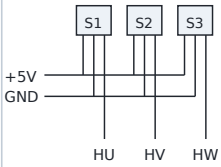
Mechanical Dimensions



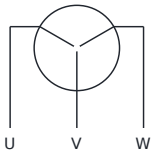
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless Motor 110 mm Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

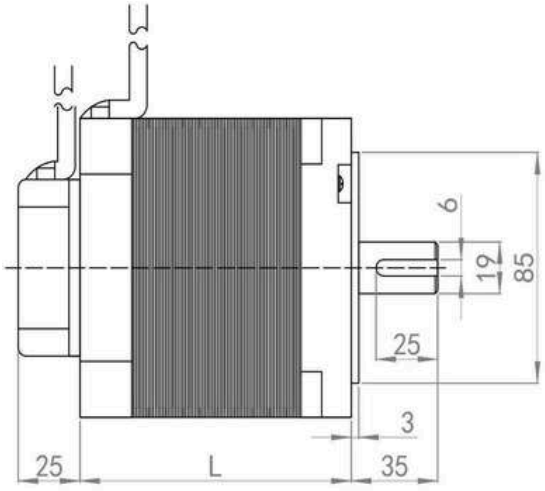
Customisable Solutions

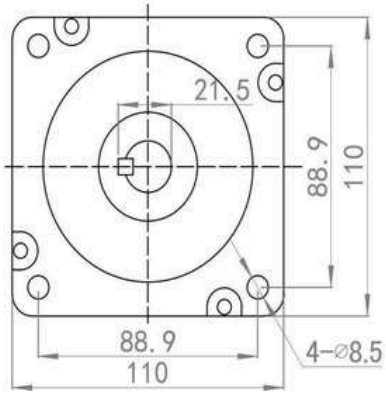
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-110BLDC-4830S	48	4000	3.26	1000	3000	3.18	26.0	9.55	78.13	110
MEC-110BLDC-31030S-1	310	4000	0.76	1500	3000	4.78	6.0	14.33	18.15	140
MEC-110BLDC-31030S-2	310	4000	1.01	2000	3000	6.37	8.1	19.10	24.19	170
MEC-110BLDC-31030S-3	310	4000	1.51	3000	3000	9.55	12.1	28.65	36.29	230

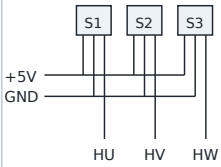
Mechanical Dimensions



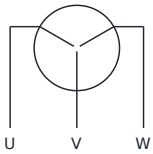


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

DC Brushless Planetary Gear Motor 110 mm Series



SPECIFICATION

Internal Gear Material	**Metal**
Bearing Type	**High-quality ball bearing**
Max. Radial Load	**≤ 700 N**
Max. Axial Load	**≤ 160 N**
Axial Play of Shaft	**≤ 0.1 mm**
Rated Permissible Torque	**160 N·m**

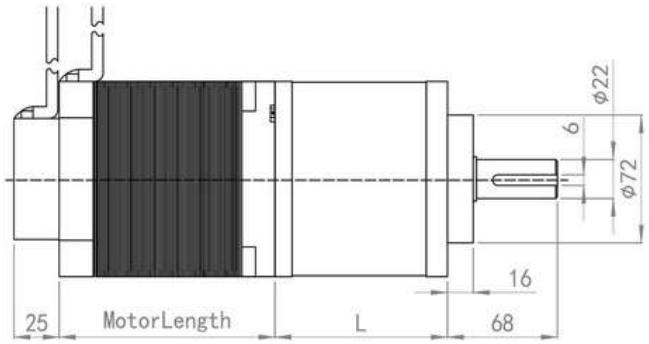
Motor Technical Data

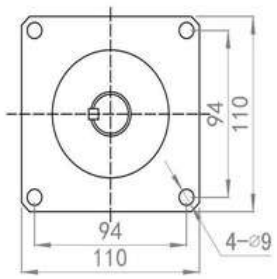
Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-110BLDC-31030S-2	310	4000	1.01	2000	3000	6.37	8.1	19.10	24.19	170
MEC-110BLDC-31030S-3	310	4000	1.51	3000	3000	9.55	12.1	28.65	36.29	230

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:4	1:10	1:24	1:36	1:64	1:96	1:144	1:216
MEC-110BLDC-31030S-3-PG110	Gearbox length	mm	117	142			167			
	No-load speed	rpm	1000	400	167	111	63	42	28	19
	Rated speed	rpm	750	300	125	83	47	31	21	14
	Rated torque	N·m	34.38	77.36	160.00	160.00	160.00	160.00	160.00	160.00

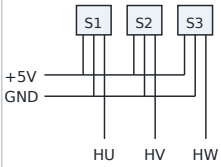
Mechanical Dimensions



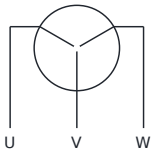


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless RV50 Worm Gear Motor 110 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**3.5 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**81 N·m**

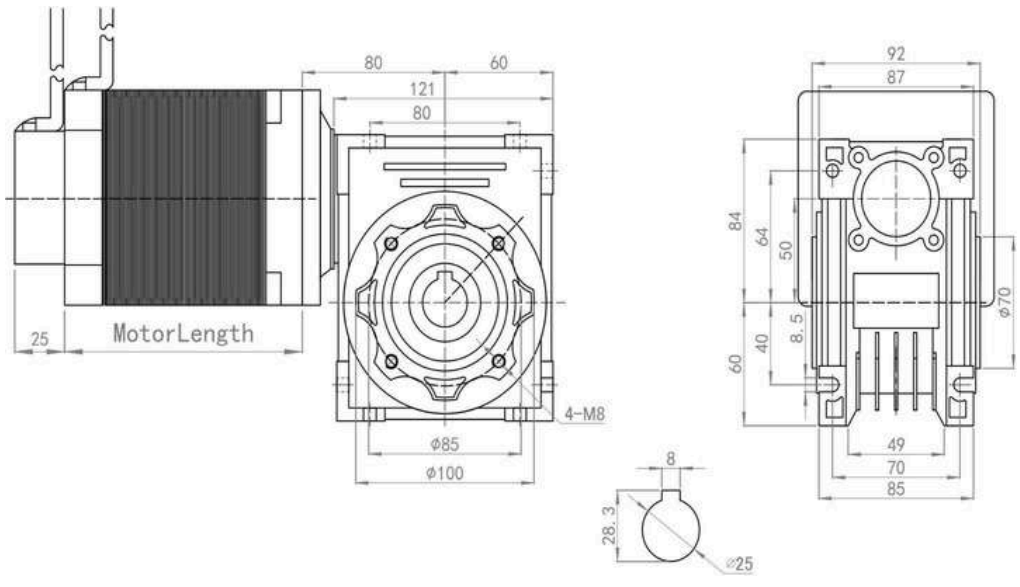
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-110BLDC-31030S-2	310	4000	1.01	2000	3000	6.37	8.1	19.10	24.19	170
MEC-110BLDC-31030S-3	310	4000	1.51	3000	3000	9.55	12.1	28.65	36.29	230

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:30	1:50	1:60	1:80	1:100
MEC-110BLDC-31030S-3-RV50	No-load speed	rpm	360	180	135	90	54	45	34	27
	Rated speed	rpm	267	133	100	67	40	33	25	20
	Rated torque	N·m	17.91	35.81	47.75	71.63	81.00	81.00	81.00	81.00

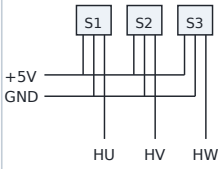
Mechanical Dimensions



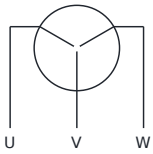
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless RV63 Worm Gear Motor 110 mm Series



SPECIFICATION

Worm Wheel Material	**Aluminium bronze**
Bearing Type	**High-quality ball bearing**
Service Life	**≥ 3000 h**
Gearbox Weight	**6.2 kg**
Ambient Temperature	**-10 °C to +40 °C**
Rated Permissible Torque	**166 N·m**

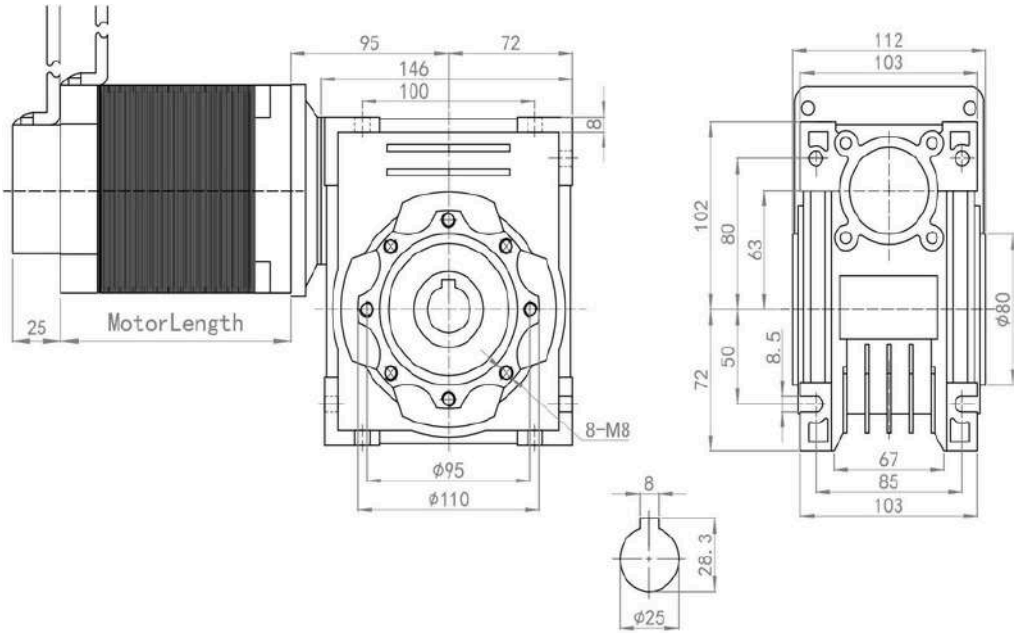
Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	W	rpm	N·m	A	N·m	A	L:mm
MEC-110BLDC-31030S-2	310	4000	1.01	2000	3000	6.37	8.1	19.10	24.19	170
MEC-110BLDC-31030S-3	310	4000	1.51	3000	3000	9.55	12.1	28.65	36.29	230

Gear Motor Technical Data

Model	Reduction ratio	1:X	1:7.5	1:15	1:20	1:30	1:50	1:60	1:80	1:100
MEC-110BLDC-31030S-3-RV63	No-load speed	rpm	360	180	135	90	54	45	34	27
	Rated speed	rpm	267	133	100	67	40	33	25	20
	Rated torque	N·m	17.91	35.81	47.75	71.63	119.38	143.25	166.00	166.00

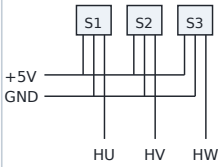
Mechanical Dimensions



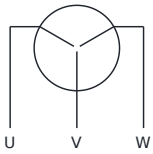
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



DC Brushless Motor QY71 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

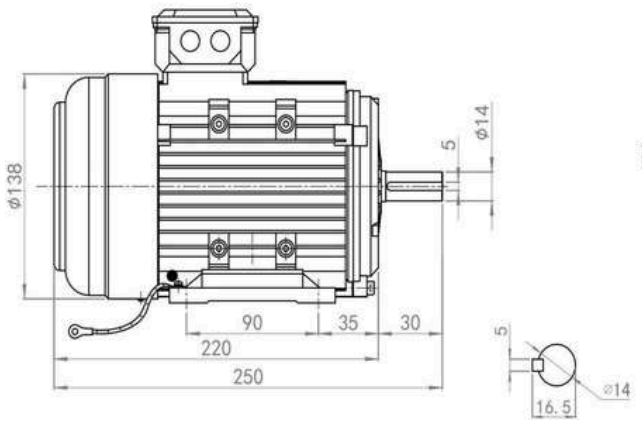
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

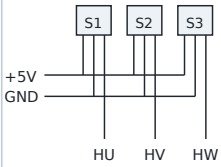
Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	kW	rpm	N·m	A	N·m	A	L:mm
MEC-71QYBLDC-2415S	24	2000	3.26	0.5	1500	3.18	26.0	9.55	78.13	220
MEC-71QYBLDC-2430S	24	4000	6.51	1	3000	3.18	52.1	9.55	156.25	220
MEC-71QYBLDC-4815S	48	2000	2.44	0.75	1500	4.78	19.5	14.33	58.59	220
MEC-71QYBLDC-4830S	48	4000	4.88	1.5	3000	4.78	39.1	14.33	117.19	220

Mechanical Dimensions

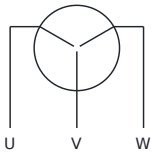


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor QY90 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

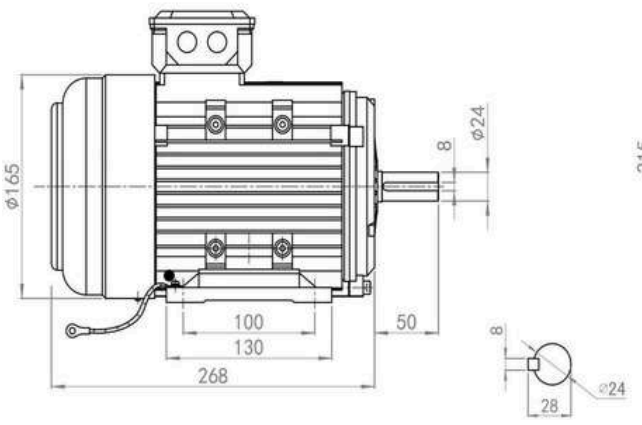
Customisable Solutions

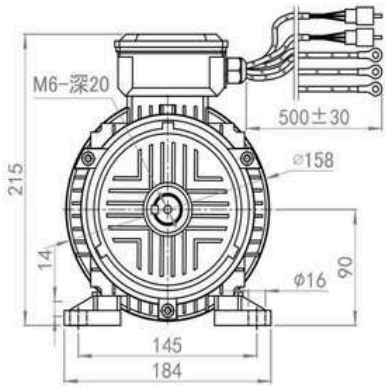
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	kW	rpm	N·m	A	N·m	A	L:mm
MEC-90QYBLDC-2415S	24	2000	7.81	1.2	1500	7.64	62.5	22.92	187.50	268
MEC-90QYBLDC-2430S	24	4000	15.63	2.4	3000	7.64	125.0	22.92	375.00	268
MEC-90QYBLDC-3615S	36	2000	5.21	1.2	1500	7.64	41.7	22.92	125.00	268
MEC-90QYBLDC-7215S	72	2000	3.26	1.5	1500	9.55	26.0	28.65	78.13	268

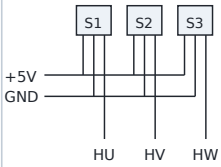
Mechanical Dimensions



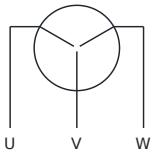


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor QY100 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

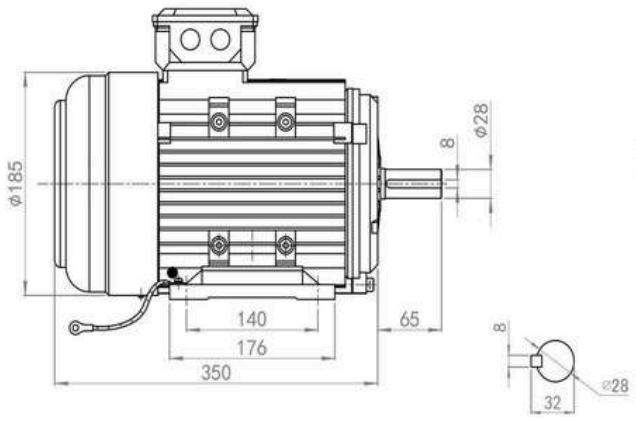
Customisable Solutions

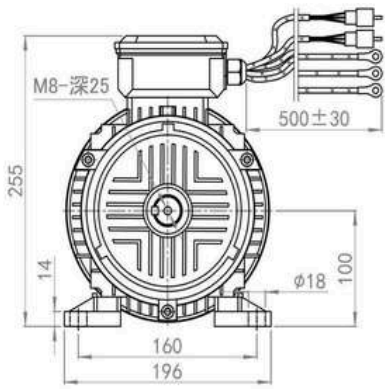
Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	kW	rpm	N·m	A	N·m	A	L:mm
MEC-100QYBLDC-2415S	24	2000	14.32	2.2	1500	14.01	114.6	42.02	343.75	350
MEC-100QYBLDC-3615S	36	2000	10.85	2.5	1500	15.92	86.8	47.75	260.42	350
MEC-100QYBLDC-4815S	48	2000	8.14	2.5	1500	15.92	65.1	47.75	195.31	350
MEC-100QYBLDC-6030S	60	4000	13.02	5	3000	15.92	104.2	47.75	312.50	350

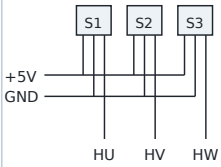
Mechanical Dimensions



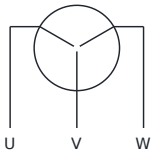


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor QY112 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

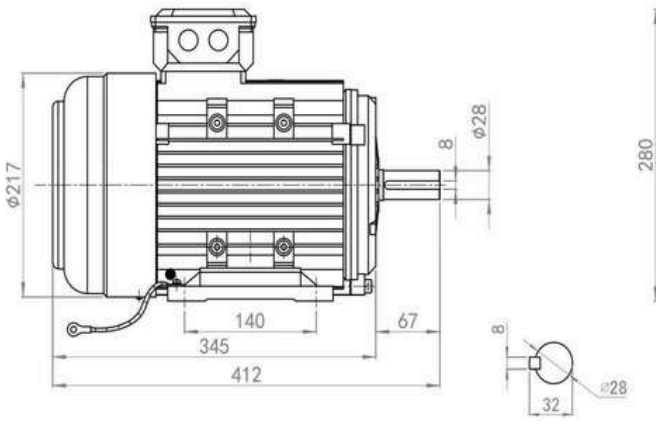
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

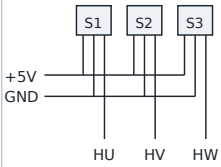
Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	kW	rpm	N·m	A	N·m	A	L:mm
MEC-112QYBLDC-4815S	48	2000	13.02	4	1500	25.47	104.2	76.40	312.50	412
MEC-112QYBLDC-6015S	48	2000	16.28	5	1500	31.83	130.2	95.50	390.63	412
MEC-112QYBLDC-6030S	60	4000	15.63	6	3000	19.10	125.0	57.30	375.00	412
MEC-112QYBLDC-7230S	72	4000	17.36	8	3000	25.47	138.9	76.40	416.67	412

Mechanical Dimensions

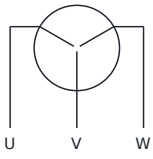


WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



All dimensions in mm. Drawings are not to scale.

www.mecwinindia.com

DC Brushless Motor QY132 Series



SPECIFICATION

Operating Environment	**-20 °C to +50 °C**
Insulation Class	**B - F**
Protection Level	**IP40**
Number of Poles	**8**
Magnet Material	**High-performance NdFeB**
Design Life	**≥ 5000 h**

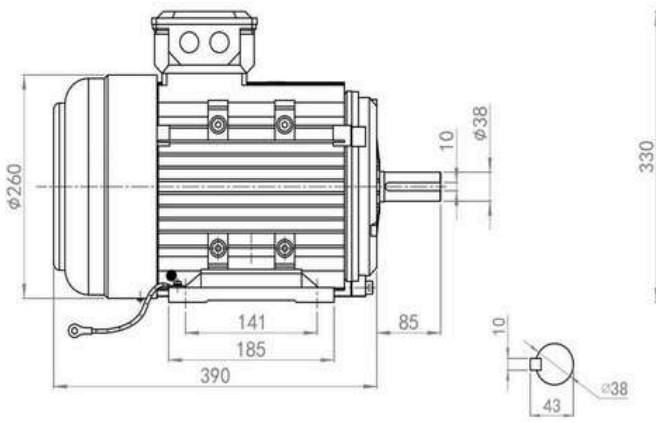
Customisable Solutions

Custom supply voltage, electrical performance and control circuitry can be provided.
Custom output shaft, mounting flange, lead wires, high-temperature insulation class, brake accessories, gearboxes, encoders and matching drives are also available.

Motor Technical Data

Model	Voltage	No-load		Rated Load				Stall		Body Length
		Speed	Current	Power	Speed	Torque	Current	Torque	Current	
	V	rpm	A	kW	rpm	N·m	A	N·m	A	L:mm
MEC-132QYBLDC-4815S	48	2000	20.51	6.3	1500	40.11	164.1	120.33	492.19	390
MEC-132QYBLDC-7215S	72	2000	16.28	7.5	1500	47.75	130.2	143.25	390.63	390
MEC-132QYBLDC-9615S	96	2000	16.28	10	1500	63.67	130.2	191.00	390.63	390
MEC-132QYBLDC-12030S	120	4000	19.53	15	3000	47.75	156.3	143.25	468.75	390

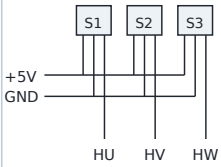
Mechanical Dimensions



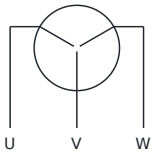
All dimensions in mm. Drawings are not to scale.

WIRING DIAGRAM

Hall Sensor Connection



Winding Connection (Star)



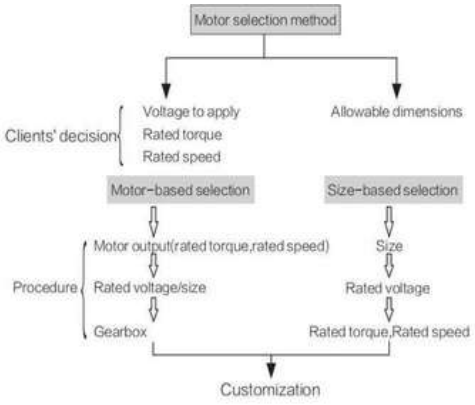
Motor Selection Method

Information Required for a Motor Proposal

Supply, control and duty conditions all influence the optimum motor solution. Please provide the following information if you would like Mecwin to recommend a motor.

- Power supply: available voltage range, rectified supply or battery, and any limitation on current.
- Accessories required: encoder, gearbox, brake or drive.
- Operating conditions: continuous or intermittent duty, direction of rotation, running time, locking and short-circuit braking.
- Environment: ambient temperature, humidity and degree of protection required.
- Dimensional limits, including gearbox, encoder and brake.

Selection Flow



Technical Terms

Term	Symbol	Unit	Term	Symbol	Unit
Voltage	U	V	Input power	P	W
Current	I	A	Output power	P	W
No-load current	I0	A	Maximum output power	Pmax	W
Stall current	Is	A	Efficiency	η	%
Speed	N	rpm	Maximum efficiency	ηmax	%
No-load speed	N0	rpm	Torque	T	mN·m, N·m
Resistance	R	Ω	Stall torque	Ts	mN·m, N·m

Technical Parameter Definitions

DC Motor Technical Parameters

No-load current - I0
The current drawn by the motor once it has reached a steady state at rated voltage with no load applied. This parameter is affected by temperature.

No-load speed - N0
The highest speed reached by the motor once it has reached a steady state at rated voltage with no load applied.

Stall torque - Ts
The torque produced by the motor at rated voltage when the output speed is zero.

$$Ts = Kt \times (U / R - I0)$$

Stall current - Is
The maximum current drawn at rated voltage when the shaft is prevented from moving, such as at start-up or when locked.

$$Is = \text{Rated voltage} / \text{Terminal resistance}$$

Maximum output power - Pmax
The maximum mechanical power the motor can deliver at rated voltage.

Maximum efficiency - ηmax
The highest ratio of mechanical output power to electrical input power.

Back-EMF constant - KE (mV/rpm)
The constant relating the voltage induced in the winding to the speed of rotation.

$$KE = 2\pi \cdot Kt / 60$$

Torque constant - Kt (g·cm/A)
The constant relating the torque developed by the motor to the current drawn.

Speed constant - Kn (rpm/V)
The change in speed per volt applied at the motor terminals under constant load.

$$Kn = 1000 / KE$$

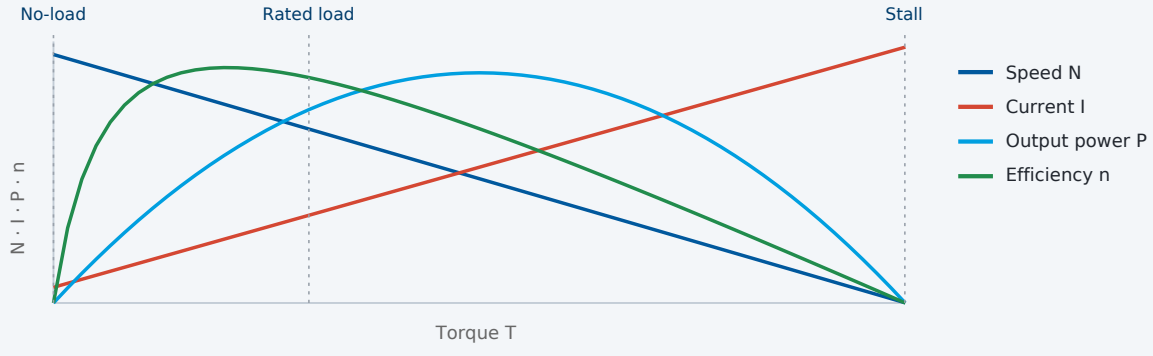
Mechanical time constant - tm (ms)
The time required for the motor to accelerate from standstill to 63% of its final no-load speed.

$$tm = R \cdot J \cdot 100 / Kt^2 \quad \text{where R = static resistance, J = rotor inertia}$$

Reading the Motor Data Tables

No-load	The motor runs at rated voltage with nothing connected to the shaft. Speed is at its highest and the current drawn is at its lowest.
Rated load	The continuous duty point at which the motor delivers its rated output power. The speed, torque and current quoted in the catalogue tables refer to this point.
Stall	The shaft is locked. Torque and current reach their maximum and no mechanical power is produced. Stall figures are reference values - continuous operation at or near stall will overheat the winding.

Typical brushless DC motor characteristic at constant supply voltage



Unit Conversion & Temperature Rise

International Torque Unit Conversion Table

Unit	ozf.in	lbf.in	lbf.ft	gf.cm	kgf.cm	kgf.m	mN·m	N·cm	N·m
1 ozf.in	1	0.0625	0.005	72	0.072	0.0007	7.062	0.706	0.007
1 lbf.in	16	1	0.083	1152.1	1.152	0.0115	113	11.3	0.113
1 lbf.ft	192	12	1	13826	13.83	0.138	1356	135.6	1.356
1 gf.cm	0.014	0.0009	0.00007	1	0.001	0.00001	0.098	0.01	0.0001
1 kgf.cm	13.89	0.868	0.072	1000	1	0.01	98.07	9.807	0.098
1 kgf.m	1389	86.8	7.233	100000	100	1	9807	980.7	9.807
1 mN·m	0.142	0.009	0.0007	10.2	0.01	0.0001	1	0.1	0.001
1 N·cm	1.416	0.088	0.007	102	0.102	0.001	10	1	0.01
1 N·m	141.6	8.851	0.783	10197	10.2	0.102	1000	100	1

Motor Temperature Rise

A motor heats up once it is energised, and the surface temperature continues to rise until it stabilises. The temperature is considered stable when it changes by no more than 2 °C over a period of three hours; the surface temperature measured at that point is taken as the final temperature of the motor, expressed in degrees Celsius (°C).

The amount by which this temperature exceeds the surrounding air (ambient) temperature is called the temperature rise, and is expressed in kelvin (K).

Rated temperature rise is the maximum permissible temperature rise of the motor winding at the specified ambient temperature (+40 °C). It is determined by the insulation class of the winding.

Calculation formula: $\theta = T2 - T1$

θ Temperature rise (K)
T1 Winding temperature in the actual cooling condition, i.e. ambient temperature (room temperature not exceeding 40 °C)
T2 Winding temperature in the heated condition

Insulation Classes & EMC

Insulation Class

The insulation class of a motor refers to the heat-resistance class of the insulating material used, graded A, E, B, F, H, C, N and R. Mecwin motors generally use class B, F or H insulation.

Temperature Class	Class A	Class E	Class B	Class F	Class H	Class C	Class N	Class R
Max. allowable temperature	105 °C	120 °C	130 °C	155 °C	180 °C	200 °C	220 °C	240 °C
Winding temperature rise limit	60 K	75 K	80 K	105 K	125 K	135 K	150 K	170 K
Performance reference temperature	80 °C	95 °C	100 °C	120 °C	145 °C	155 °C	170 °C	190 °C

EMC - Suppression of Electrical Noise

When a DC brush motor rotates, spark current occurs as the commutator switches. This spark can become electrical noise and disturb the control circuit. Such noise is reduced by connecting a capacitor across the motor.

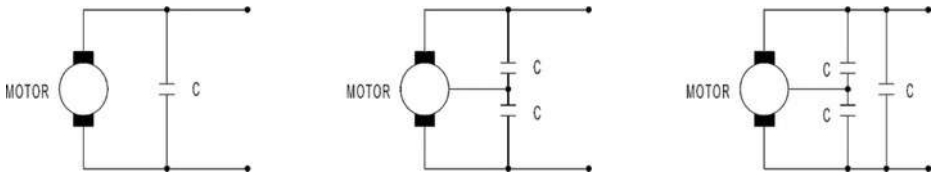
Capacitors and chokes can be fitted at the motor terminals to reduce electrical noise. The most effective way of eliminating the spark is to fit the component on the rotor close to the source, although this method is expensive.

1. Fit noise-suppression components inside the motor - varistor (D/V), annular capacitor, rubber ring resistance (RRR) or chip capacitor. Effective against high-frequency noise.

2. Fit noise-suppression components outside the motor - external capacitor (electrolytic or ceramic type) or choke. Effective against low-frequency noise.

Methods 1 and 2 may be used separately; using both together gives the best noise-reduction result.

Connection examples



C = capacitance (capacitor) L = inductance (choke)

Degree of Protection – IP Code

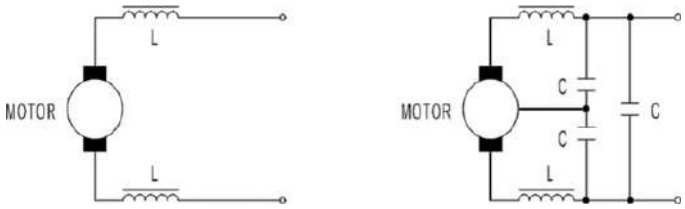
The IP code to IEC 60529 states the degree of protection provided by the motor enclosure. The first digit describes protection against solid objects and the second digit protection against water. Catalogue motors are supplied to IP40 as standard - protected against solid objects of 1 mm and above, with no protection against water. Please contact Mecwin for applications requiring a higher degree of protection.

1st digit	Protection against solid objects
0	No protection
1	Objects of 50 mm and above
2	Objects of 12.5 mm and above
3	Objects of 2.5 mm and above
4	Objects of 1 mm and above
5	Dust protected
6	Dust tight

2nd digit	Protection against water
0	No protection
1	Vertically falling drops
2	Falling drops, enclosure tilted 15 degrees
3	Spraying water
4	Splashing water
5	Water jets
6	Powerful water jets
7	Temporary immersion

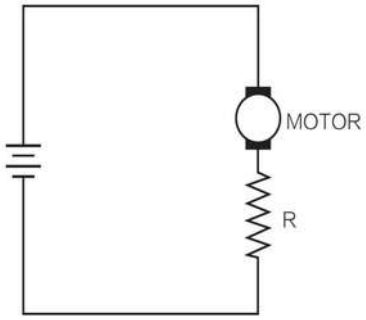
Motor Control Methods

Noise Suppression with Chokes



1. Control by Resistance

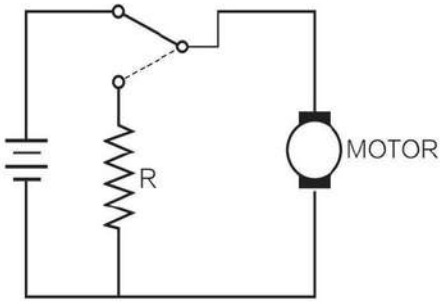
A resistor is placed in series between the power supply and the DC motor. Increasing the resistance limits the current and controls the speed of the motor. Using a variable resistance changes the linear characteristic of the motor.



Series resistance controls the motor current and speed

2. Motor Braking (Short-circuit Braking)

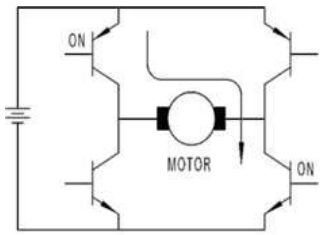
The motor terminals are forced into a short circuit to stop the motor. This shortens the braking time and provides powerful braking control. Maximum braking is obtained when R = 0.



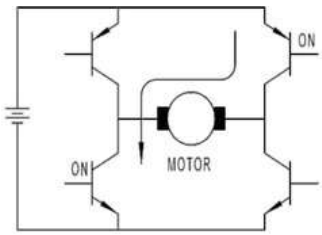
Maximum braking effect is obtained when R = 0

3. Bidirectional Driving by Transistors

Bidirectional motor drive is achieved by switching an H-bridge circuit built from NPN and PNP transistors.



Forward run - upper left and lower right transistors on



Reverse run - lower left and upper right transistors on

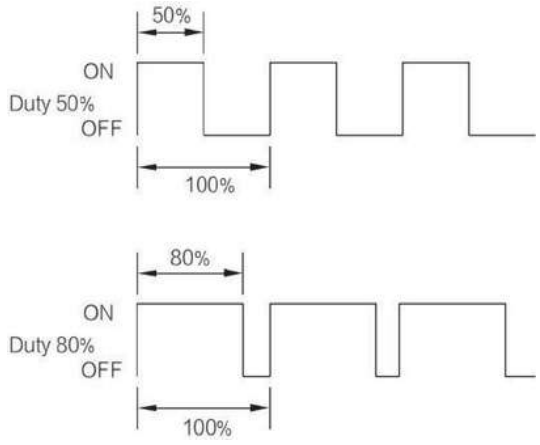
PWM Control

Pulsewidthmodulationmethod

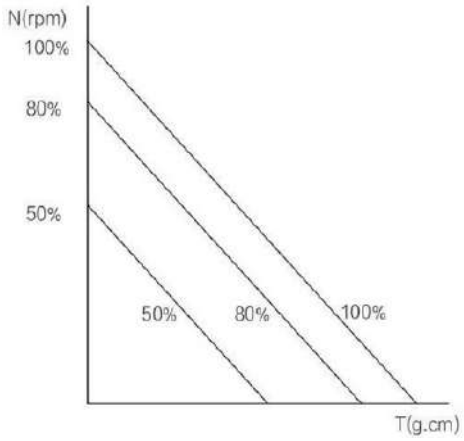
Principle of PWM Control

The basic principle of PWM control is to keep the switching period fixed and vary the ratio of ON time to OFF time of the pulse. This varies the average energy delivered to the motor and thereby controls its speed.

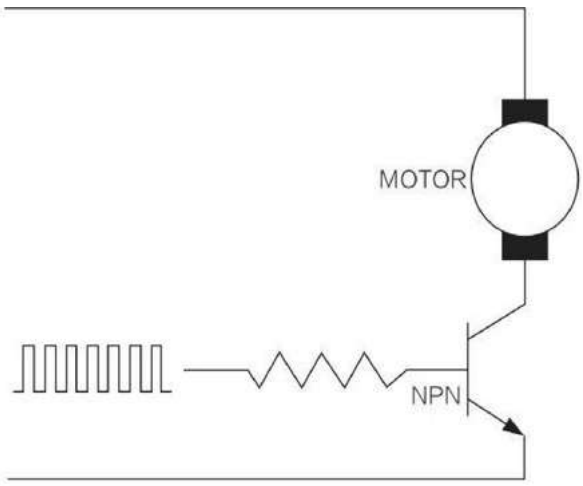
Duty ratio



Speed / torque characteristic



Reference Circuit



PWM control reference circuit diagram

Application Notes

- The average voltage applied to the motor is the supply voltage multiplied by the duty ratio, so speed varies approximately in proportion to the duty ratio at constant load.
- A higher switching frequency reduces audible noise and current ripple, but increases switching losses in the drive.
- Brushless DC motors in this catalogue are driven by a three-phase inverter with hall-sensor commutation; the same duty-ratio principle applies to each phase.
- Matching Mecwin drives and controllers are available for every motor in this catalogue - please contact our sales team with the supply voltage, control signal and duty cycle required.

Motor and Gearbox Combinations

Gearboxes

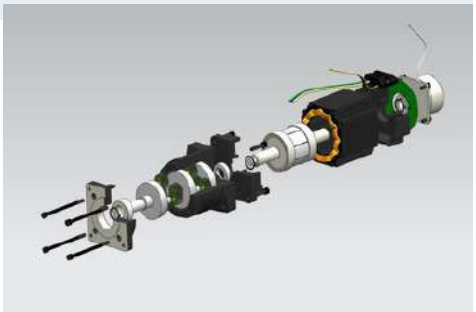
A gearbox is a transmission mechanism whose construction reduces the motor speed and increases the output torque. Output speed and torque can be calculated approximately using the formulae below. Reduction ratios are set individually for each gearbox type.



Characteristics of Gearbox Types

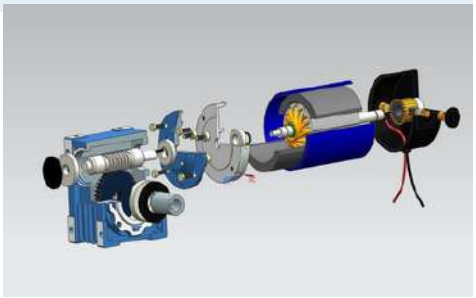
Planetary Gearbox

- Transmits high torque
- Efficiency of 91% per stage
- Input and output on the same centre line
- Input and output rotate in the same direction



RV Worm Gearbox

- Low noise and smooth running; suitable for long-term continuous duty in harsh environments
- Wide ratio range - 12 reduction ratios from 5:1 to 100:1 for each frame
- Generous cooling ribs give the housing excellent heat conduction
- Ratios of 80:1 and 100:1 are effectively self-locking, which can save separate braking equipment where braking is required



Gearbox Range Overview

Planetary	Rated torque	Ratios	Pages
PG28	2 N·m	1:4 - 1:139	6, 7
PG36	3 N·m	1:4 - 1:139	9, 10
PG42	6 - 9 N·m	1:4 - 1:216	13, 14, 15
PG56	30 N·m	1:3.6 - 1:121	17
PG80	45 - 60 N·m	1:4 - 1:216	21, 27
PG86	80 N·m	1:4 - 1:216	29
PG110	160 N·m	1:4 - 1:216	36

Worm	Rated torque	Ratios	Pages
RV25	15 N·m	1:7.5 - 1:60	18, 22
RV30	24 N·m	1:7.5 - 1:60	19, 23, 30
RV40	53 N·m	1:7.5 - 1:100	31
RV50	81 N·m	1:7.5 - 1:100	32, 37
RV63	166 N·m	1:7.5 - 1:100	38

Troubleshooting – DC Brake

DC Brake Troubleshooting

Fault	Probable Cause	Remedy
Brake does not work or works poorly	Power is not supplied	Supply the power
	The lining is worn out	Replace with a new lining
	Insufficient or incorrect supply voltage	Supply the correct voltage
	Power supply unit damaged	Replace with a new power supply unit
	Intrusion of foreign matter	Clean the parts
	Connection has fallen off	Connect again
	Lining has got stuck	Clean the parts
	Braking coil burnt out	Replace with a new braking coil
	Clearance too large	Adjust the clearance
	The lining is contaminated with oil	Clean the lining
	Overloaded	Re-design the load
	Brake surface deflected	Replace the parts
	Descending load too large	Re-design the mechanism
	Wrong model selected	Choose the correct model
	Ambient temperature too high	Improve the ambient temperature

If the fault cannot be cleared using the table above, please contact Mecwin with the motor and brake model numbers and details of the application

Spur Gearbox & Gear Efficiency

Spur Gearbox

- Torque capability is comparatively low, but a thin and quiet design is possible
- Efficiency of 79% per stage
- Input and output may be on the same or on different centre lines
- Direction of output rotation depends on the number of gear stages



Gear Efficiency

A gear reducer has some mechanical loss, caused by gear meshing and bearing friction. As a general rule, efficiency is calculated per stage of gear mesh as 79% for a spur gear reducer, 91% for a planetary gear reducer and 55% for an RV gear reducer. The actual figure varies considerably with motor torque, speed and other conditions, so these values should be used for reference only. Efficiency falls significantly when the gearbox is used in a low-torque condition.

Planetary gearbox, 3 stages
$\eta = 0.91 \times 0.91 \times 0.91 \approx 0.753$ (75.3%)

Spur gearbox, 3 stages
$\eta = 0.79 \times 0.79 \times 0.79 \approx 0.493$ (49.3%)

Speed and Torque Relationship

Gear motor speed = motor speed (rpm) / reduction ratio

Gearbox output torque = motor output torque (Ts) × reduction ratio × gear efficiency (η)

Example: motor speed 6000 rpm, rated torque T = 0.3 N·m, fitted with a 100:1 gearbox of 60% transmission efficiency.
Gear motor speed = 6000 rpm / 100 = 60 rpm
Gearbox output torque = 0.3 N·m × 100 × 60% = 18 N·m

Selecting a Geared Motor - Worked Example

Requirement: 40 rpm continuous output speed at 6 N·m, 48 V DC supply.

1. Find the required output speed in the Gear Motor Technical Data tables. On page 15, MEC-42BLDC-4840S-1-PG42 gives a rated output speed of 40 rpm at a ratio of 1:100.
2. Check the rated torque in the same column: 9.00 N·m, which exceeds the 6 N·m required.
3. Check the rated permissible torque of the gearbox (9 N·m for this combination) and the permissible radial and axial loads on the output shaft.
4. Confirm the supply voltage and the mechanical envelope from the dimension drawing, then add any options required - electromagnetic brake (M) or encoder (BFE).

Resulting part number: **MEC-42BLDC-4840S-PG42-100K**

Troubleshooting – Gear Reducer

Gear Reducer Troubleshooting

Symptom	Fault	Probable Cause	Remedy
Noise	Gear knocking	Gear surface damaged	Replace the damaged gear set
	Continuous noise	Bearing damaged	Replace the damaged bearing(s)
	Periodic noise	Foreign matter attached to gear surface	Check the gear tooth surfaces
	Hissing	Insufficient oil	Add lubricating oil
	Intermittent noise	Lubricating oil is unclean	Replace with new lubricating oil
Vibration	Fixed base vibration	Mounting plane out of true	Re-adjust the fixed base
	Output shaft vibration	Bearing damaged	Replace the damaged bearing(s)
	Internal gear part vibration	Gear damaged	Replace the damaged gear
	Body vibration	Gear set incorrectly installed	Re-adjust the gear set
Oil leakage	Oil seal leakage	Oil seal hardened	Replace the damaged oil seal
	Body leakage	Sand hole in the housing	Replace the housing
	Joint surface leakage	O-ring damaged	Replace the damaged O-ring
Overheating	Oil seal	Oil seal over-tightened	Replace the over-tightened oil seal
	Overheated body	Overloaded	Re-calculate the load horsepower
	Lack of lubricating oil	Insufficient oil	Add lubricating oil

Note: permissible motor temperature rise is within 80 °C to the applicable national standard.

Installation and Handling Notes

- Mount the motor on a flat, rigid surface and check alignment with the driven machine before tightening the fixings.
- Do not strike the shaft when fitting couplings, pulleys or sprockets - use the tapped hole in the shaft end where one is provided.
- Keep radial and axial shaft loads within the limits stated on the relevant product page; geared motors are particularly sensitive to overhung loads.
- Allow free air circulation around the motor. Catalogue ratings assume an ambient temperature of -20 °C to +50 °C for motors and -10 °C to +40 °C for worm gearboxes.
- Check the direction of rotation and the hall-sensor sequence with the motor uncoupled before connecting it to the load.
- Store in a dry place and avoid condensation. Protect the shaft and flange from corrosion during prolonged storage.

MECWIN Global Customers



International Compliance Support

- Products can be designed and validated for applicable:
- IEC and EN standards
- CE conformity requirements
- UL and CSA requirements
- RoHS and REACH compliance
- ATEX requirements for hazardous-area applications
- MIL-STD environmental testing requirements
- Customer-specific testing and qualification standards

Certifications and third-party testing can be undertaken based on product application, destination country and customer requirements.

Partner with Mecwin

From Agriculture to Aviation, Mecwin delivers engineered motor solutions for global applications. Design in India. Manufacture in Bengaluru. Supply to the World.

CONTACT US



Our Service & Support team are directly accessible in 18+ states in India.
MECWIN's trained service engineers can help with site survey, system sizing, installation, commissioning and on time service support.



Mecwin Technologies India Pvt.Ltd

Manufacturing Plant
Mecwin Infinity Tech Park, #42/2, Opp. Machohalli Lake, K.G. Lakkenahalli Road, Machohalli Village, Dasanapura-2 Hobli, Bengaluru North Taluk, 560091, India

Overseas Office
12 Amber Close, Rugby CV22 5GE, United Kingdom.

Customer Care No : + 91 97412 29797

Email : marketing@mecwinindia.com

Scan to Reach Us